The Impact of Gaming on Mental Health:

Exploring Anxiety, Life Satisfaction, and Social Phobia Among Gamers

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Introduction

Video games have grown from a niche pastime into one of the largest forms of entertainment on the planet. What began in the 1970s with simple arcade cabinets like Pong and Space Invaders has expanded into a global industry that earns more revenue each year than film and music combined. The hobby is no longer limited to teenagers in basements. Today, more than three billion people across every continent play video games on phones, consoles, computers, and handheld devices. The variety of games available has exploded as well, from quick puzzle apps that fill a five-minute break to sprawling multiplayer worlds that can occupy hundreds of hours. Gaming has become a regular part of daily life for a huge slice of the global population. With so many people involved, the way that gaming connects to other parts of life, including health, school, work, and relationships, has become a serious subject of public interest. Understanding those connections matters because the choices made by individual players, parents, schools, and governments all depend on having good information about what gaming actually does to people.

Gaming touches every demographic group. Players range from young children who are first learning to use a controller, all the way up to retirees who use games to stay mentally sharp or to keep in touch with grandchildren. Industry surveys show that the average gamer in the United States is in their mid-thirties, and the gender split is closer to even than many people assume. Mobile gaming in particular has brought in millions of casual players who would not describe themselves as gamers at all. Players come from every country, every income level, and every background, which is part of what makes the topic so important to study at a broad scale. Because gaming is such a common activity, even small effects on health or behavior can add up to large effects across the whole population. Schools, healthcare providers, employers, and families are all affected by how the people in their lives spend time with games. Anyone who interacts with children, teenagers, or young adults today is interacting with people whose lives include gaming in some form.

As gaming has grown, so have concerns about how it might affect mental health. Researchers, doctors, journalists, and parents have all raised questions about whether long hours of play can cause anxiety, depression, or social withdrawal in some people. Specific worries that come up again and again include gaming addiction, sleep deprivation from late-night sessions, loss of motivation for school or work, and isolation from family and friends. The World Health Organization added Gaming Disorder to its International Classification of Diseases in 2018, which gave a formal medical name to a pattern of behavior that had been discussed informally for years. At the same time, news stories about extreme cases of unhealthy gaming have shaped public perception, sometimes in ways that go beyond what the actual research supports. The result is a

public conversation that often swings between treating gaming as a harmless hobby and treating it as a serious mental health risk. Sorting through these competing views requires looking carefully at real evidence rather than relying on the loudest opinions.

The relationship between gaming and mental health is more complicated than headlines often suggest. Some studies have linked heavy gaming with higher rates of anxiety, social phobia, and lower life satisfaction, especially in young people. Other studies have found exactly the opposite, showing that gaming can help reduce stress, build friendships across long distances, improve hand-eye coordination, and support cognitive skills like problem-solving and quick decision-making. Multiplayer games in particular have given many people, including those with social anxiety, a low-pressure way to interact with others. The same game can be a healthy outlet for one player and a harmful escape for another, depending on how it fits into the rest of that person’s life. Personality, age, family situation, the type of game being played, whether play is solo or social, and the total amount of time spent all influence how gaming and mental health interact. There is no single answer that applies to every player, and most serious researchers now agree that the question must be studied with that complexity in mind.

Because the stakes are high and the picture is so mixed, gaming and mental health is a topic that deserves careful attention from many different groups. Families want guidance on how much gaming is too much for their kids and on warning signs to watch for. Schools and universities want to understand how gaming habits relate to academic performance and student well-being. Healthcare providers want practical advice they can give to patients who report problems related to gaming. Game developers themselves have a stake in understanding how their products affect the people who use them, and a growing number of studios now build features designed to encourage healthier play patterns. Policymakers face questions about advertising, age ratings, microtransactions, and screen-time limits, and they need solid information to make good decisions. Even individual players benefit from a clearer view of how their habits might affect their mood, sleep, social life, and long-term mental health. The pages that follow examine real data from gamers around the world to look for patterns that can help inform all of these audiences.

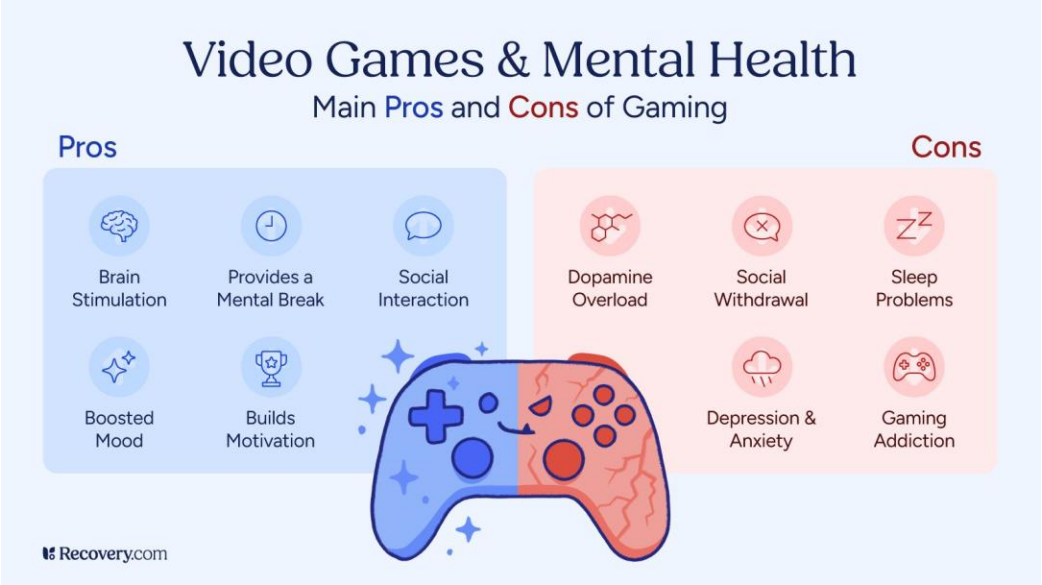


Figure 1: An overview of the main pros and cons of gaming on mental health (Recovery.com, https://recovery.com/resources/video-games-and-depression/).

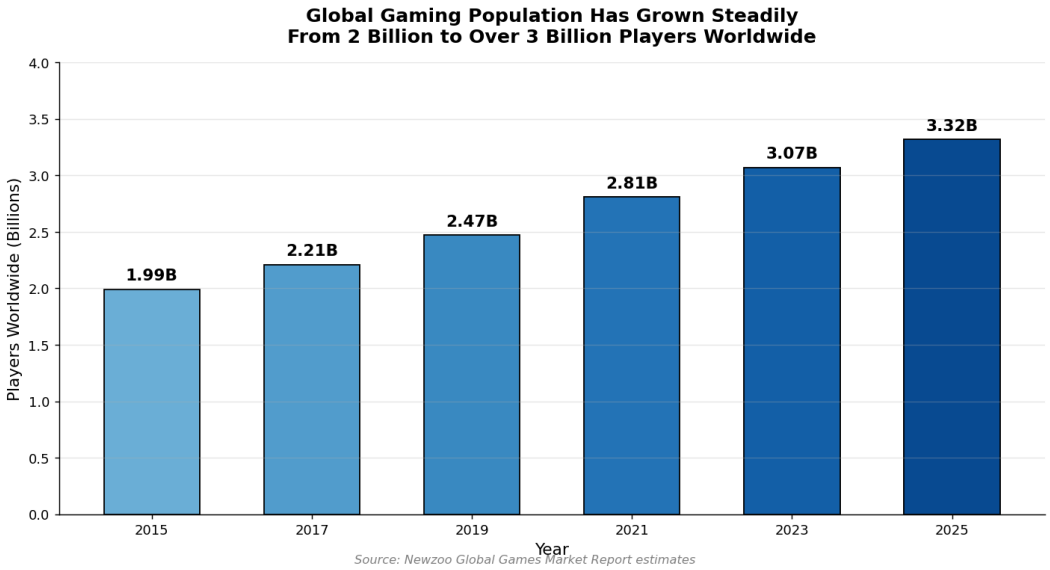


Figure 1B: The global gaming population has grown steadily over the past decade, reaching more than three billion players worldwide (Source: Newzoo Global Games Market Report estimates).

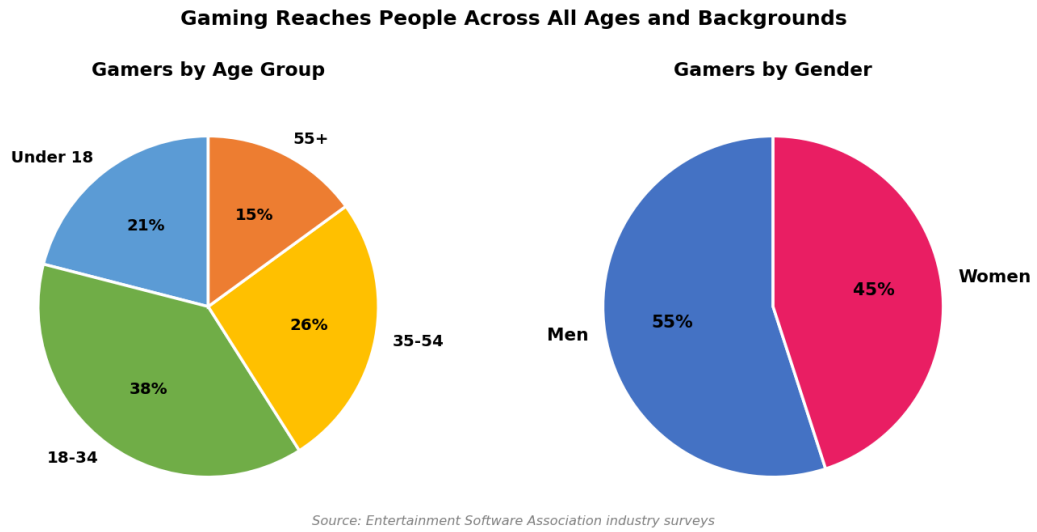


Figure 1C: Players come from every age group and gender, which makes the impact of gaming a question that touches almost every part of society (Source: Entertainment Software Association industry surveys).

Personal Relevance

I chose this topic because gaming has been a big part of my life for as long as I can remember. I have spent hundreds of hours playing MMORPGs like Final Fantasy XIV and Guild Wars 2, along with story-driven games like NieR: Automata and the Persona series. Through all of that, I have seen how gaming can be a great way to relax but also how it can lead to staying up way too late or skipping out on other things. I am pretty introverted, so gaming has always been my go-to way to wind down after a long day of school or work. The online multiplayer side of things can be fun and social, but it can also feel like a lot sometimes. This project lets me take something I actually care about and combine it with the data science skills I am learning in this class. I am curious to see what the data says about how gaming and mental health are connected.

Research Questions

Below are ten questions that will help guide this project. These questions might change a little as the project moves forward and as results come in from different machine learning models.

1. Is there a statistically significant correlation between weekly gaming hours and generalized anxiety disorder (GAD-7) scores?
2. Do gamers who play primarily in multiplayer modes exhibit different levels of social phobia compared to those who play singleplayer games?
3. Can machine learning models accurately predict a player's anxiety level (Low, Mild, Moderate, Severe) based on their gaming habits and demographic information?
4. How does life satisfaction (as measured by the SWL scale) differ across gaming platforms (PC, Console, Mobile)?
5. Is there a relationship between the amount of time spent watching gaming streams and social phobia (SPIN) scores?
6. Do players who report higher narcissism scores tend to have different anxiety or life satisfaction profiles compared to those with lower narcissism?
7. What game genres are associated with the highest and lowest levels of player engagement, and does engagement level relate to session duration or player level?
8. Can clustering (unsupervised learning) reveal distinct player profiles based on a combination of gaming behavior and mental health metrics?
9. Does the age of a player moderate the relationship between gaming hours and anxiety levels?
10. What combination of features (gaming hours, platform, genre, demographics) best predicts player engagement level using supervised classification models?

Data Gathering

Two labeled datasets were collected for this project. Both of them relate to the topic of gaming and mental health, but they look at it from different angles. The first dataset is based on survey responses about anxiety, life satisfaction, and social phobia from gamers. The second dataset focuses more on gaming behavior like how long people play, what genres they prefer, and how engaged they are. Together, these two datasets give a solid base for running both supervised and unsupervised machine learning models later in the class.

Dataset 1: Online Gaming Anxiety Study

This dataset comes from a worldwide survey done by Marian Sauter and Dejan Draschkow in 2017. The survey looked at how gaming habits connect to mental health. It used well-known psychological tests including the GAD-7 for anxiety, the Satisfaction with Life Scale (SWL), and the Social Phobia Inventory (SPIN). People who took the survey also shared basic info about themselves and how they play games.

Source: https://www.kaggle.com/datasets/divyansh22/online-gaming-anxiety-data

Nature of Data: Mixed (quantitative and qualitative). It has numeric scores like the GAD items, SWL items, SPIN items, Hours, Age, Narcissism, and streams. It also has categorical columns like Game, Platform, Gender, Work, Degree, and GADE.

Label: A new label called "Anxiety_Level" was created from the GAD_T total score. It has four categories: Low (0-4), Mild (5-9), Moderate (10-14), and Severe (15-21). The original GADE column also works as a label with four categories that describe how difficult anxiety makes daily life.

Dimensions: Raw: 13,464 rows and 55 columns. After cleaning: 11,233 rows and 43 columns (42 variables + 1 engineered label).

Dataset 1: Online Gaming Anxiety Study (Raw Data Sample)

GAD1	GAD2	GAD3	GAD_T	SWL_T	SPIN_T	Game	Platform	Hours	Gender	Age
0	0	0	1	23	5.0	Skyrim	Console (PS, Xbox, ..	15.0	Male	25
1	2	2	8	16	33.0	Other	PC	8.0	Male	41
0	2	2	8	17	31.0	Other	PC	0.0	Female	32
0	0	0	0	17	11.0	Other	PC	20.0	Male	28
2	1	2	14	14	13.0	Other	Console (PS, Xbox, ..	20.0	Male	19
0	0	0	1	17	13.0	Other	Console (PS, Xbox, ..	4.0	Male	24
0	0	0	0	16	26.0	Other	PC	30.0	Male	29
0	0	0	0	23	nan	Other	Console (PS, Xbox, ..	25.0	Female	23

Figure 2: Raw data sample from Dataset 1 showing selected columns.

Dataset 2: Predict Online Gaming Behavior

This dataset was put together by Rabie El Kharoua in 2024. It looks at gaming from a behavior standpoint rather than a mental health one. It includes things like player age, gender, location, how long they play, what genre they prefer, whether they make in-game purchases, and how engaged they are. This dataset pairs well with Dataset 1 because it focuses on what players actually do in games rather than how they feel, which gives a more complete picture.

Source: https://www.kaggle.com/datasets/rabieelkharoua/predict-online-gaming-behavior-dataset

Nature of Data: Mixed (quantitative and qualitative). The numeric columns are Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, AchievementsUnlocked, and InGamePurchases. The categorical columns are Gender, Location, GameGenre, and GameDifficulty.

Label: "EngagementLevel" with three categories: Low, Medium, and High.

Dimensions: Raw: 40,034 rows and 13 columns. A random sample of 2,000 rows was taken for this project, resulting in 2,000 rows and 12 columns (after dropping PlayerID).

Dataset 2: Predict Online Gaming Behavior (Raw Data Sample)

PlayerID	Age	Gender	Location	GameGenre	PlayTimeHours	InGamePurchases	GameDifficulty	SessionsPerWeek	AvgSessionDurationMin	PlayerLevel	AchievementsUnlocke	EngagementLevel
9000	43	Male	Other	Strategy	16.271118760553215	0	Medium	6	108	79	25	Medium
9001	29	Female	USA	Strategy	5.5259613805705 66	0	Medium	5	144	11	10	Medium
9002	22	Female	USA	Sports	8.223755243499511	0	Easy	16	142	35	41	High
9003	35	Male	USA	Action	5.265351277318268	1	Easy	9	85	57	47	Medium
9004	33	Male	Europe	Action	15.53194452113429	0	Medium	2	131	95	37	Medium
9005	37	Male	Europe	RPG	20.561855414112557	0	Easy	2	81	74	22	Low
9006	25	Male	USA	Action	9.752716365932256	0	Hard	1	50	13	2	Low
9007	25	Female	Asia	RPG	4.401729344841462	0	Medium	10	48	27	23	Medium

Figure 3: Raw data sample from Dataset 2 showing all columns.

Data Cleaning and Preparation

Cleaning Dataset 1: Online Gaming Anxiety Study

Cleaning Dataset 1 took a few steps because it is a large dataset with a lot of missing values. First, 13 columns that were not useful for the analysis were removed. These included things like the row number (S. No.), Timestamp, free-text fields like earnings and whyplay, and geographic codes. The highestleague column was completely empty so it got dropped too. Next, any rows that were missing a value in the GADE column were removed since that is the original label and incomplete responses are not helpful. For the SPIN columns and other numeric fields like Hours, streams, and Narcissism, missing values were filled in using the median of each column. This keeps the overall spread of the data about the same. After that, any remaining rows with missing values in columns like Work and Degree were dropped.

A new label called "Anxiety_Level" was created using the GAD_T total score. The GAD-7 test has standard score ranges, so scores from 0 to 4 were labeled "Low," 5 to 9 as "Mild," 10 to 14 as "Moderate," and 15 to 21 as "Severe." This gives a label with four categories that can be used for classification models later on. Finally, 44 rows were removed where people said they gamed more than 80 hours per week, which does not make sense since there are not that many waking hours in a week. After all of that, the cleaned dataset has 11,233 rows and 43 columns.

Dataset 1: Cleaned Data Sample

GAD_T	SWL_T	SPIN_T	Hours	streams	Narcissism	Gender	Age	Platform	Anxiety_Level
1	23	5.0	15.0	0.0	1.0	Male	25	Console (PS, Xbox, ...	Low
8	16	33.0	8.0	2.0	1.0	Male	41	PC	Mild
8	17	31.0	0.0	0.0	4.0	Female	32	PC	Mild
0	17	11.0	20.0	5.0	2.0	Male	28	PC	Low
14	14	13.0	20.0	1.0	1.0	Male	19	Console (PS, Xbox, ...	Moderate
1	17	13.0	4.0	0.0	2.0	Male	24	Console (PS, Xbox, ...	Low
0	16	26.0	30.0	8.0	2.0	Male	29	PC	Low
12	12	55.0	2.0	0.0	1.0	Female	23	Console (PS, Xbox, ...	Moderate

Figure 4: Cleaned Dataset 1 sample showing key variables and the engineered Anxiety_Level label.

Cleaning Dataset 2: Predict Online Gaming Behavior

Dataset 2 did not need much cleaning since it was already in good shape with no missing values or duplicate rows. The main thing that was done was taking a random sample of 2,000 rows from the original 40,034 to keep the size manageable. The PlayerID column was also dropped because it is just an ID number and does not help with analysis. The EngagementLevel label was checked and it has three categories (Low, Medium, High) with a decent spread across all three. The final cleaned dataset has 2,000 rows and 12 columns, with 7 numeric and 4 categorical columns plus the label.

Dataset 2: Cleaned Data Sample (2000 Row Sample)

Age	Gender	Location	GameGenre	PlayTimeHours	InGamePurchases	GameDifficulty	SessionsPerWeek	AvgSessionDurationMinu	PlayerLevel	AchievementsUnlocke	EngagementLevel
49	Male	Other	Action	3.326911320481104	1	Medium	16	63	43	6	Medium
29	Female	USA	Action	19.62646247850453	0	Hard	10	99	48	11	Medium
32	Male	Asia	Sports	23.32208579857633	0	Medium	11	17	12	34	Medium
49	Male	Asia	Action	10.348791401534967	0	Medium	7	149	44	38	Medium
48	Female	Other	Simulation	6.238741796428113	0	Easy	10	127	14	4	Medium
43	Female	Asia	Strategy	3.8458676093275033	0	Easy	11	170	60	34	High
15	Male	Asia	Action	12.496477746473357	0	Medium	5	11	67	40	Low
18	Male	Europe	Simulation	10.850420835012434	0	Hard	16	85	84	25	High

Figure 5: Cleaned Dataset 2 sample (2,000 row random sample).

Data Visualization

Dataset 1 Visualizations

The charts below show what is going on inside the gaming anxiety survey data. The first chart shows how many people fall into each anxiety level. Most of the respondents landed in the "Low" category, and fewer people showed up in Mild, Moderate, and Severe. This makes sense because most people in any group tend to have lower anxiety scores.

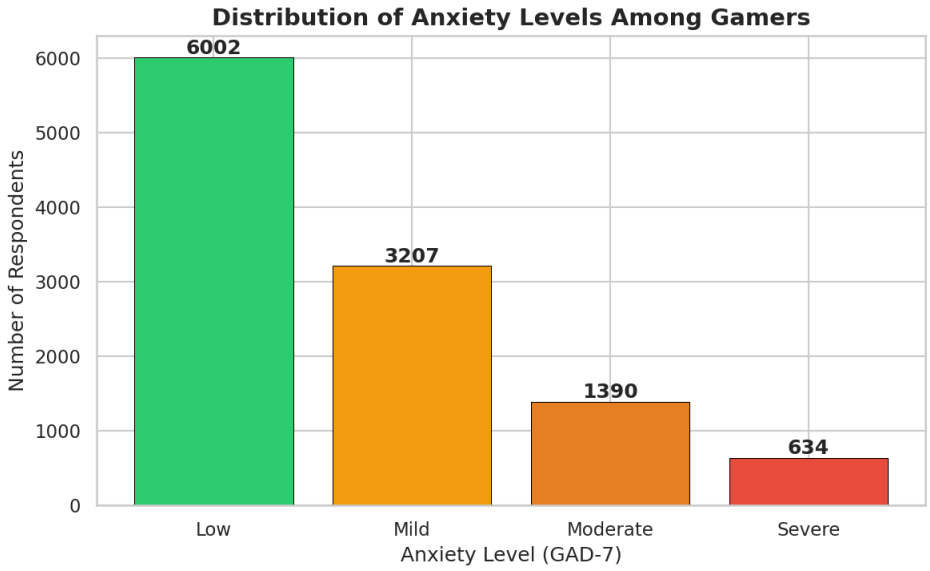


Figure 6: Distribution of anxiety levels among gaming survey respondents.

This scatter plot looks at whether there is a visible link between how many hours someone plays per week and their anxiety score. The dots are spread out pretty evenly, which suggests that the relationship is not as simple as saying more gaming means more anxiety. Higher anxiety scores show up at all levels of play time, not just among heavy gamers.

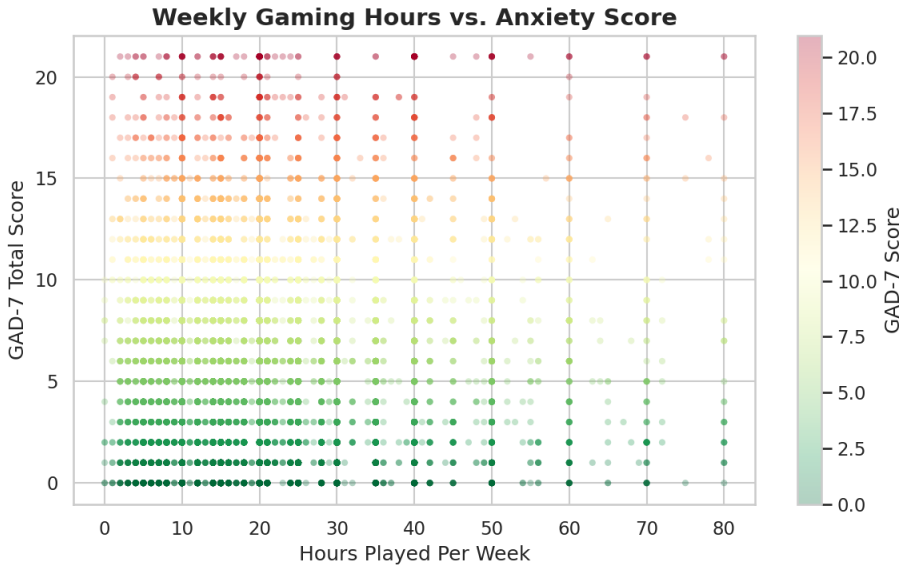


Figure 7: Scatter plot of weekly gaming hours versus GAD-7 anxiety score.

The heatmap below shows how the main numeric variables relate to each other. Some interesting things stand out. Anxiety and social phobia have a moderate positive connection, meaning people with higher anxiety also tend to have higher social phobia scores. Anxiety and life satisfaction have a negative connection, so higher anxiety generally means lower life satisfaction. Gaming hours do not have a strong connection to any of the mental health scores, which is worth noting.

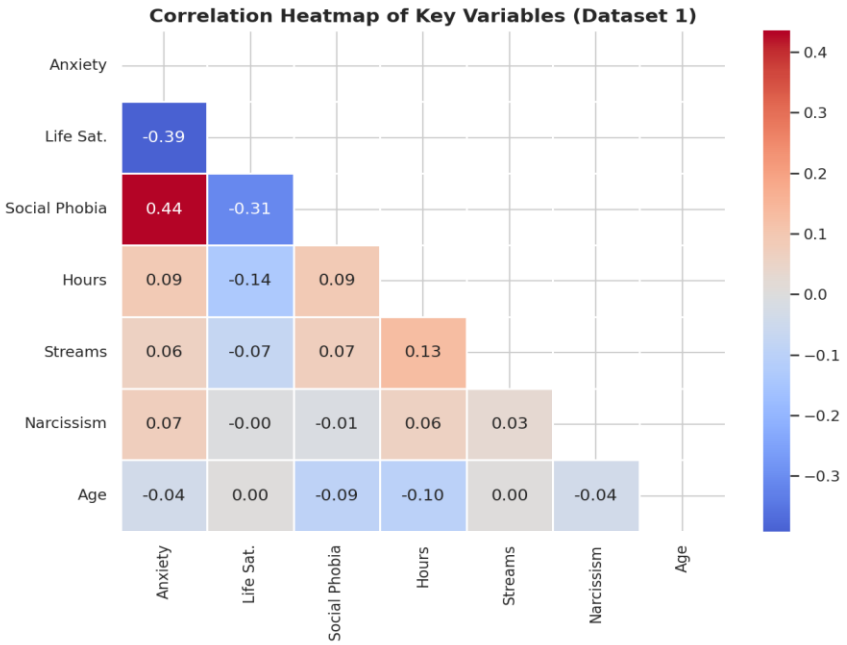


Figure 8: Correlation heatmap of key numeric variables in Dataset 1.

This boxplot compares anxiety scores across different gaming platforms. PC, Console, and Smartphone/Tablet users all show pretty similar anxiety distributions, though there are small differences in the medians. This will be looked at more closely with statistical tests and machine learning models hopefully at a later date.

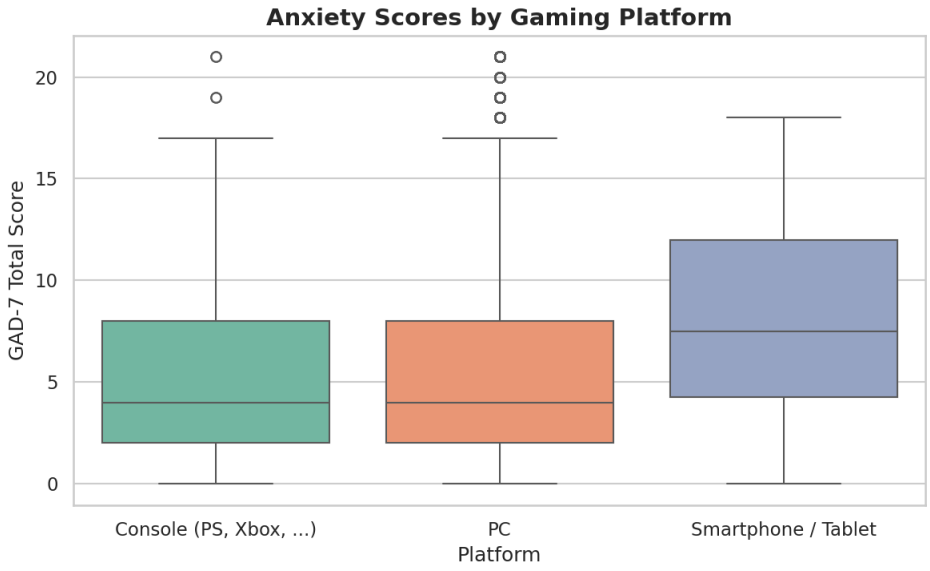


Figure 9: Boxplot of GAD-7 anxiety scores by gaming platform.

Dataset 2 Visualizations

The engagement level chart for Dataset 2 shows that about half of the sampled players fall into the "Medium" category, while "High" and "Low" are roughly equal. This is a good sign for classification tasks, as all three groups have sufficient data to work with.

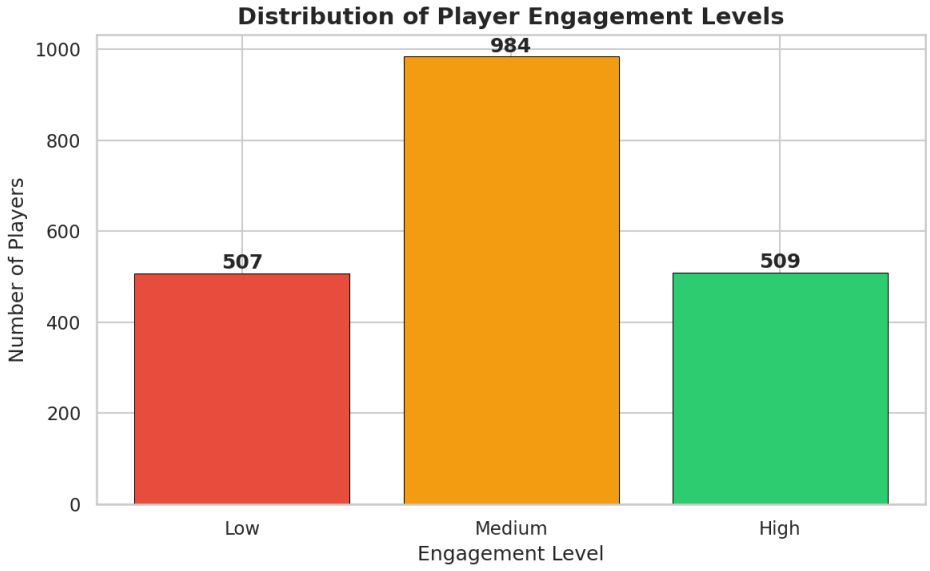


Figure 10: Distribution of player engagement levels in Dataset 2.

The boxplot of play time by game genre shows that all five genres (Action, RPG, Simulation, Sports, Strategy) have similar play time distributions, with medians around 10 to 12 hours. This means play time does not change much by genre in this dataset. Other things, like how often someone plays or how long each session lasts, might matter more.

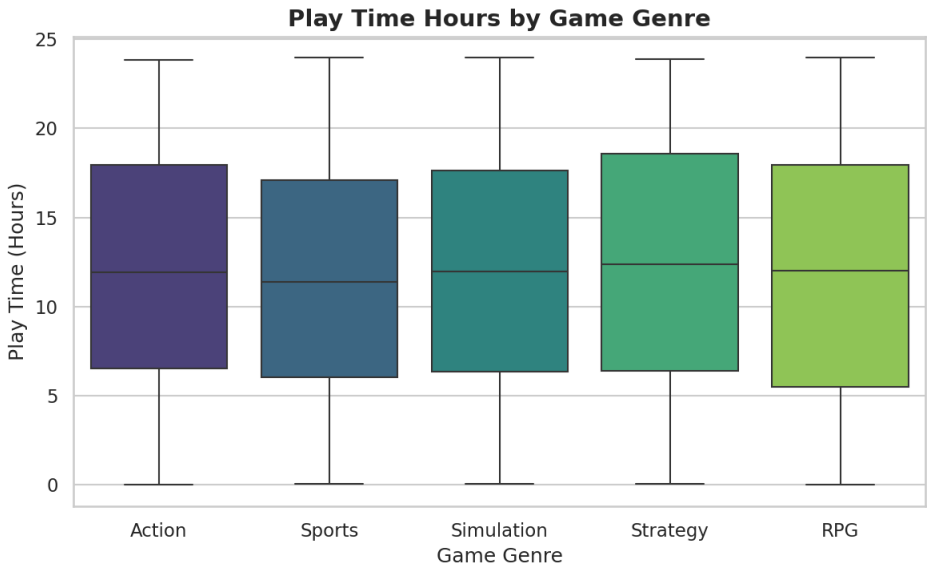


Figure 11: Play time hours by game genre.

This scatter plot shows session duration versus player level, with each dot colored by engagement level. There is a clear pattern here. Players with high engagement tend to be at higher levels and play for longer sessions, while low-engagement players cluster in the bottom-left area with shorter sessions and lower levels. This separation is a good sign that these features could work well as predictors in classification models.

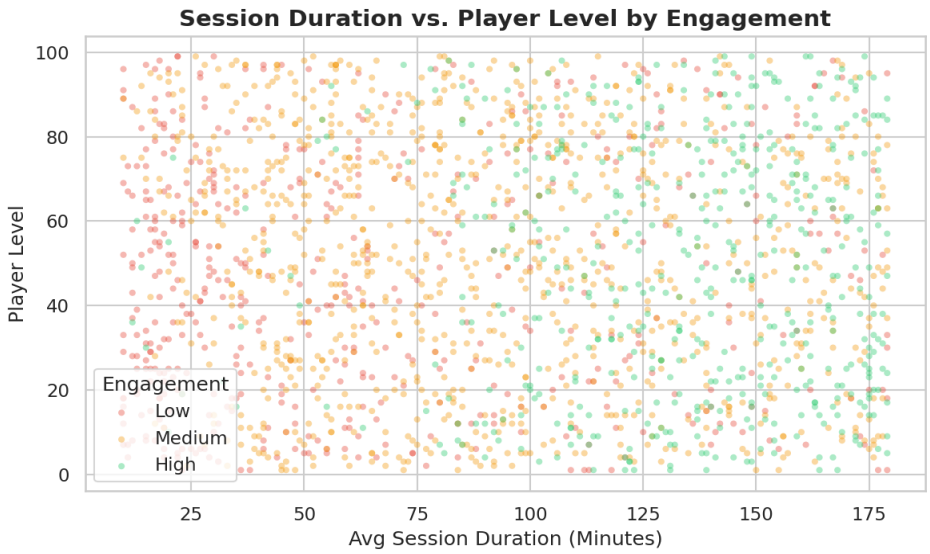


Figure 12: Session duration vs. player level, colored by engagement level.

The heatmap for Dataset 2 shows that most of the numeric columns do not have strong connections with each other. That is actually a good thing for modeling because it means the variables are each bringing something different to the table. The strongest link is between player level and achievements unlocked, which makes sense since higher-level players have had more time to unlock things.

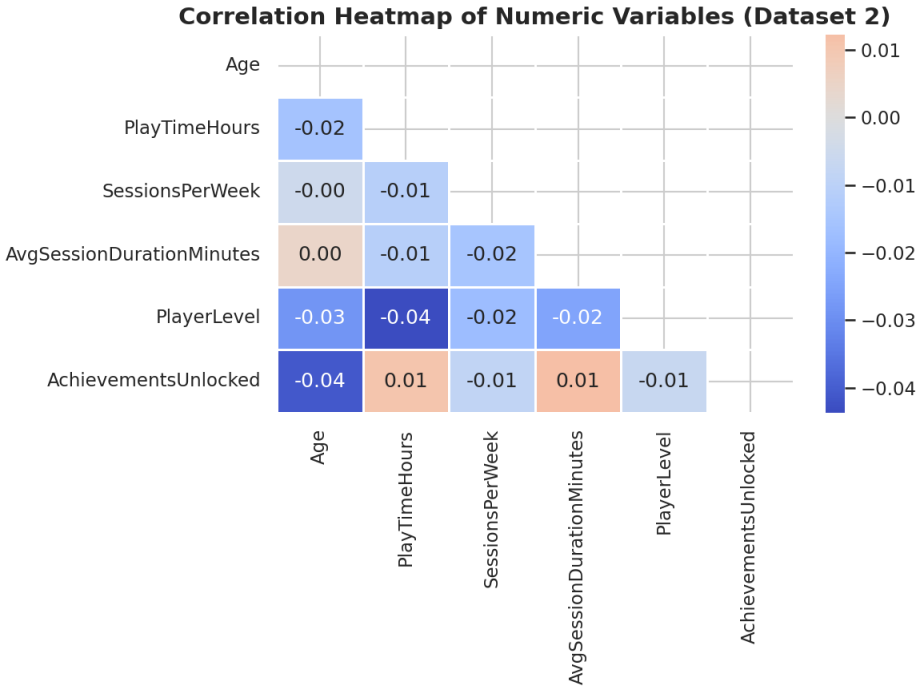


Figure 13: Correlation heatmap of numeric variables in Dataset 2.

Clustering and PCA

This section applies unsupervised machine learning methods to the Online Gaming Behavior dataset (Dataset 2). The goal is to see if clustering can reveal distinct player profiles based on gaming behavior, without using the EngagementLevel label. Three approaches are used: KMeans clustering on the full feature space, KMeans after reducing the data to three dimensions with PCA, and Agglomerative Hierarchical Clustering. A random sample of 2,000 players was used with six quantitative features: Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, and AchievementsUnlocked.

Before clustering, all six features were normalized using Min-Max scaling so that each one falls between 0 and 1. This is important because KMeans uses distance to assign points to clusters, and without normalization a feature like PlayerLevel (range 1 to 99) would dominate a feature like PlayTimeHours (range 0 to 24). The images below show the data before and after normalization.

Before Normalization (First 10 Rows)

Age	PlayTimeHours	SessionsPerWeek	AvgSessionDurationMinutes	PlayerLevel	AchievementsUnlocked
49.0	3.33	16.0	63.0	43.0	6.0
29.0	19.63	10.0	99.0	48.0	11.0
32.0	23.32	11.0	17.0	12.0	34.0
49.0	10.35	7.0	149.0	44.0	38.0
48.0	6.24	10.0	127.0	14.0	4.0
43.0	3.85	11.0	170.0	60.0	34.0
15.0	12.5	5.0	11.0	67.0	40.0
18.0	10.85	16.0	85.0	84.0	25.0
45.0	15.88	13.0	175.0	25.0	18.0
44.0	16.72	4.0	53.0	8.0	30.0

Figure 14: Data before Min-Max normalization (first 10 rows).

After Min-Max Normalization (First 10 Rows)

Age	PlayTimeHours	SessionsPerWeek	AvgSessionDurationMinutes	PlayerLevel	AchievementsUnlocked
1.0	0.1385	0.8421	0.3136	0.4286	0.1224
0.4118	0.8192	0.5263	0.5266	0.4796	0.2245
0.5	0.9735	0.5789	0.0414	0.1122	0.6939
1.0	0.4317	0.3684	0.8225	0.4388	0.7755
0.9706	0.2601	0.5263	0.6923	0.1327	0.0816
0.8235	0.1602	0.5789	0.9467	0.602	0.6939
0.0	0.5214	0.2632	0.0059	0.6735	0.8163
0.0882	0.4527	0.8421	0.4438	0.8469	0.5102
0.8824	0.6629	0.6842	0.9763	0.2449	0.3673
0.8529	0.6977	0.2105	0.2544	0.0714	0.6122

Figure 15: Data after Min-Max normalization, scaled to 0-1 range.

KMeans without PCA

KMeans clustering was applied to the normalized 6D data using sklearn's KMeans library. Three different values of k were tested: k=2, k=3, and k=4. The elbow method was also used to visualize the relationship between k and inertia (within-cluster sum of squares) for k=2 through k=7.

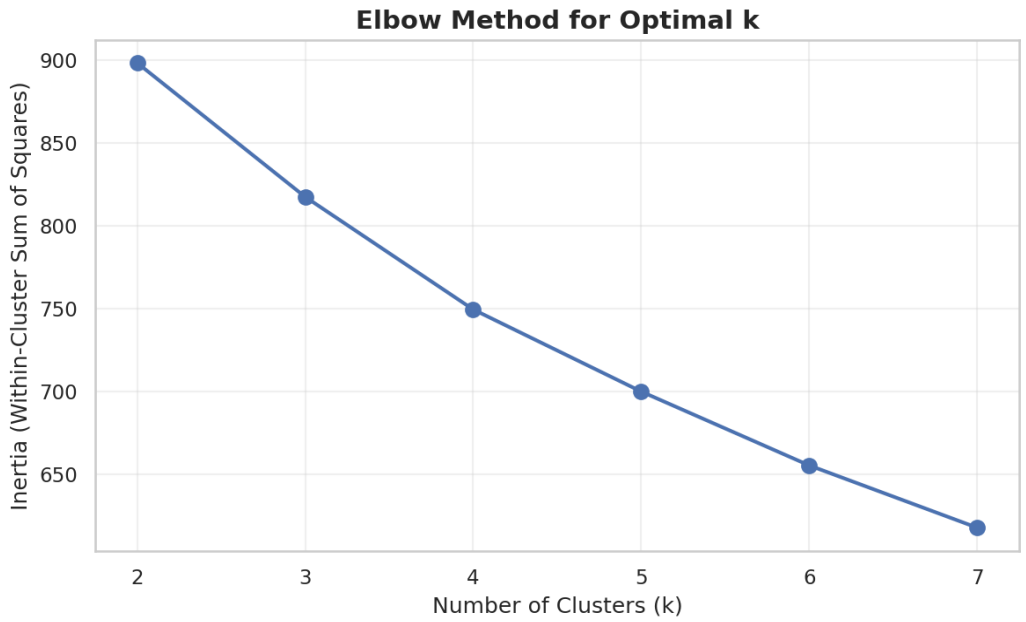


Figure 16: Elbow method plot showing inertia for different values of k.

For k=2, the silhouette score was 0.1271 and the two clusters were roughly even in size (969 and 1,031 players). For k=3, the silhouette score dropped slightly to 0.1144 with cluster sizes of 741, 669, and 590. For k=4, the silhouette score came back up to 0.1251 with four clusters of similar size (around 480 to 530 each). None of the silhouette scores are particularly high, which suggests that the data does not have very strong natural groupings in the full 6D space. This makes sense given that the features are relatively independent of each other, as the correlation heatmap in Module 1 showed.

The parallel coordinates plot below shows the k=2 result. Each thin line represents a player, and the thick lines show the cluster centroids. The two clusters mainly differ in their Age, SessionsPerWeek, and PlayerLevel values. One cluster captures younger, higher-level, more frequent players while the other captures older players with fewer sessions.

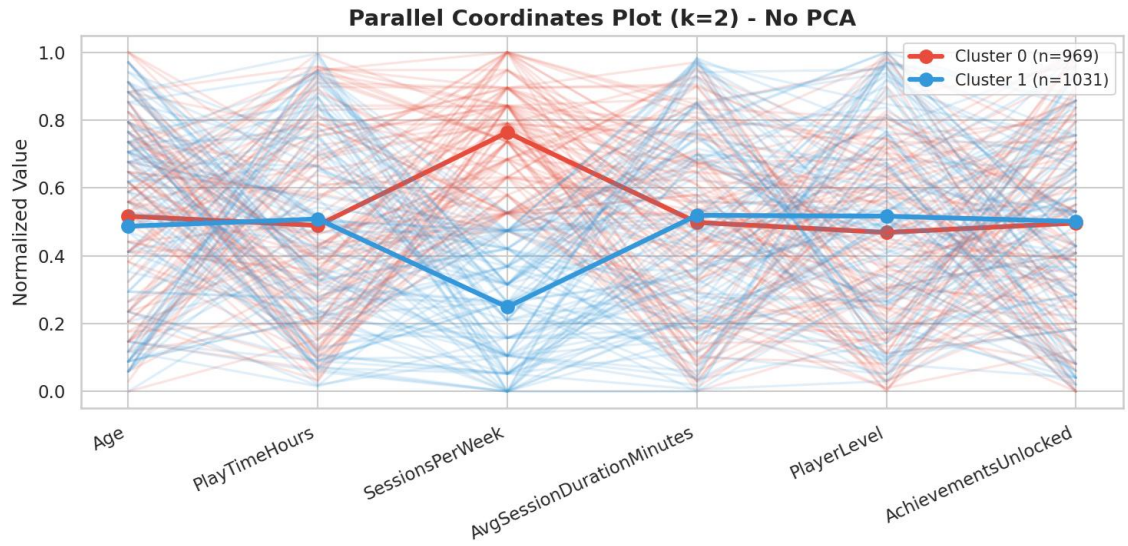


Figure 17: Parallel coordinates plot for KMeans with k=2.

The pairplot below visualizes the k=3 clustering across four of the six features. Each scatterplot shows a different pair of variables, and the points are colored by cluster. The clusters overlap quite a bit in most pairs, but there is some separation visible in certain combinations like Age vs. PlayerLevel and PlayTimeHours vs. SessionsPerWeek.

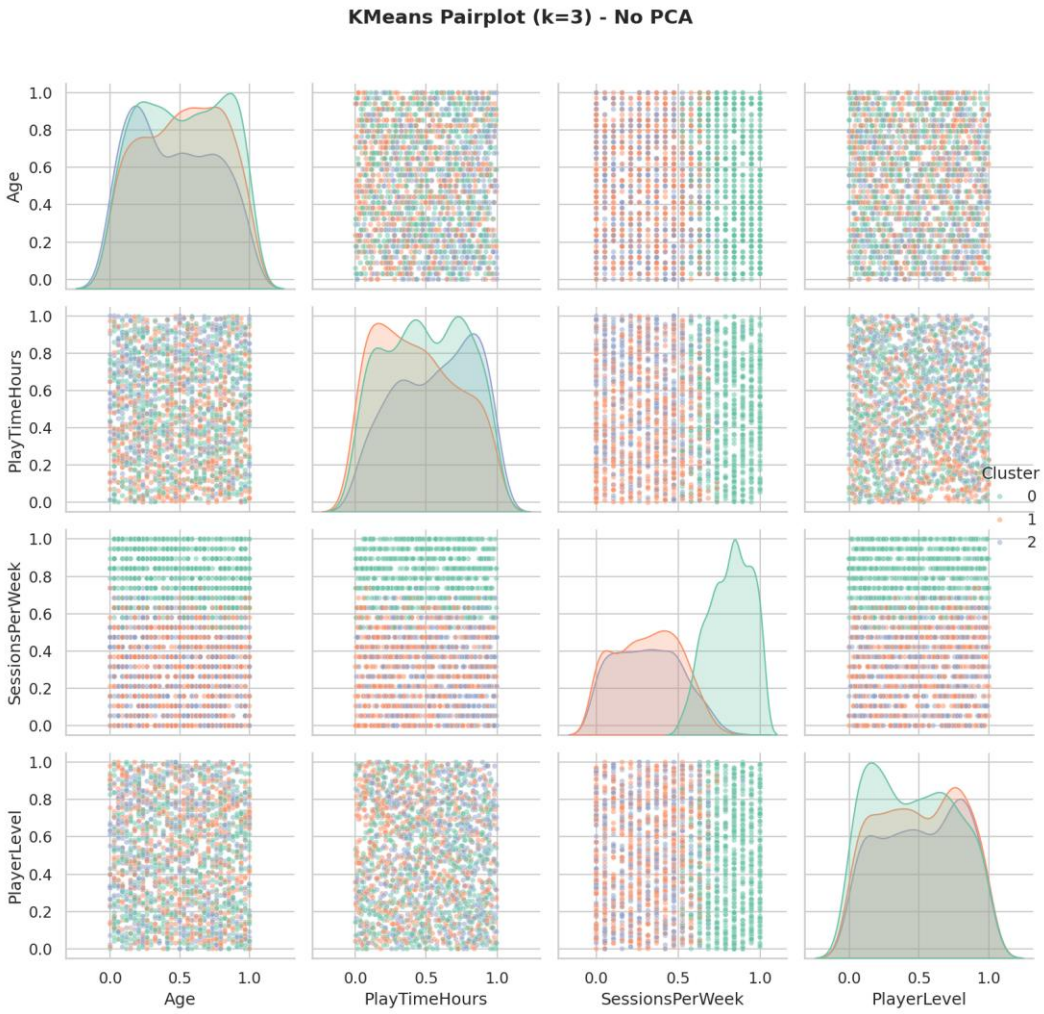


Figure 18: Pairplot for KMeans with k=3 showing cluster separation across features.

The radar plots below show the centroid of each cluster for k=4. Each axis represents one of the six features, and the filled area shows the average values for that cluster. This makes it easy to see how the clusters differ. For example, Cluster 0 has high Age and SessionsPerWeek but low PlayerLevel, while Cluster 2 has high PlayTimeHours and AchievementsUnlocked but lower Age. These profiles suggest that KMeans is picking up on different types of gamers even without using labels.



Figure 19: Radar plots of cluster centroids for KMeans with k=4.

Overall, the KMeans results without PCA show that there are some differences between player groups, but they are not super strong. The relatively low silhouette scores tell us that the clusters are not tightly separated. This could be because the data is spread out pretty evenly across the six features, or because the real structure in the data is not well captured by distance in the full 6D space. The radar plots for k=4 do reveal some interesting player profiles though, like a cluster of younger, high-achievement players versus a cluster of older, frequent but lower-level players. These patterns connect to Research Question 8, which asked whether clustering could reveal distinct player profiles based on gaming behavior.

KMeans Using PCA

PCA (Principal Component Analysis) was used to reduce the 6D normalized data down to 3 dimensions before running KMeans. The three principal components explained about 52.4% of the total variance (PC1: 17.7%, PC2: 17.4%, PC3: 17.3%). The explained variance is spread fairly evenly across the three components, which means no single component dominates. This is consistent with the low correlations between features that were noted in Module 1.

PCA-Reduced Data (First 10 Rows, 3 Components)

PC1	PC2	PC3
0.7269	-0.1522	0.1888
0.106	-0.0272	-0.0313
0.0391	0.2802	-0.3136
-0.009	-0.249	-0.3358
0.5694	-0.3951	-0.2496
0.0499	-0.0999	-0.0854
-0.5604	0.2454	0.3234
-0.0727	0.4266	0.4336
0.3721	-0.1151	-0.4568
0.0425	-0.2436	-0.4259

Figure 20: PCA-reduced data showing 3 principal components (first 10 rows).

KMeans was run on the PCA-reduced data with k=2, k=3, and k=4. The silhouette scores improved across the board compared to the non-PCA results. For k=2 the score was 0.2384, for k=3 it was 0.2301, and for k=4 it was 0.2505. The best silhouette score came from k=4 with a score of 0.2505, which means the data separates most cleanly into four groups after PCA. This makes sense because reducing the dimensionality removes noise and helps KMeans focus on the most important patterns.

The 3D scatter plot below shows the clustering result for k=4 in the PCA-reduced space. Each point is a player and the colors represent the four clusters. The clusters are more visually distinct than they were in the pairplots of the original 6D data, which confirms that PCA helped clarify the groupings.

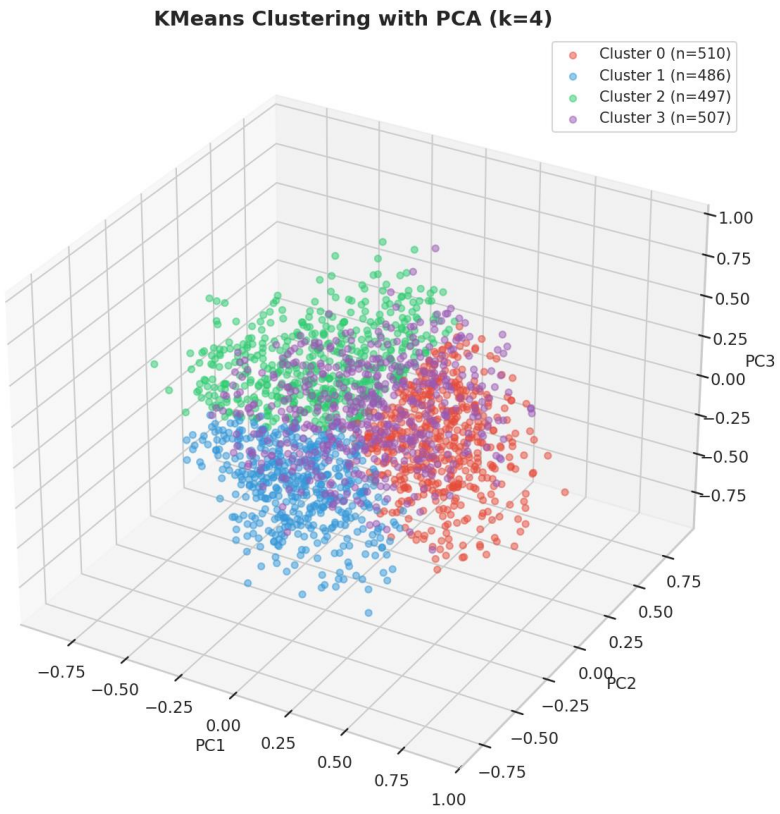


Figure 21: 3D visualization of KMeans clustering (k=4) after PCA.

Comparing the PCA and non-PCA results side by side, the PCA-based clustering had consistently higher silhouette scores for every value of k. This suggests that reducing the data to 3D removed some noise and made the cluster structure easier to detect. The best overall result was k=4 with PCA, scoring 0.2505 compared to 0.1251 without PCA. While these scores are still moderate, the improvement is meaningful and shows that PCA is a useful preprocessing step for clustering this type of data.

In terms of what this means for the project, the PCA-based clusters support the idea that players can be grouped into distinct profiles even without knowing their engagement level. The four clusters likely represent different combinations of gaming intensity, experience level, and session patterns. This finding ties directly into Research Question 8 about using unsupervised methods to discover player types, and could also help with Research Question 10 about which features best predict engagement.

Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering was performed using sklearn's AgglomerativeClustering library with Ward linkage on a subset of 200 samples from the normalized data. Ward linkage was chosen because it minimizes the total within-cluster variance, which is similar to what KMeans does and makes the two methods easier to compare. A subset of 200 samples was used because dendrograms become unreadable with thousands of data points.

The dendrogram below shows the hierarchical structure of the data. The y-axis represents the distance (Ward linkage distance) at which clusters merge. The red dashed line shows the approximate cut threshold that produces three clusters. Looking at the dendrogram, there are a few main branches that split early on, but the sub-branches merge at similar heights, which again reflects the fact that the data does not have super strong natural groupings.

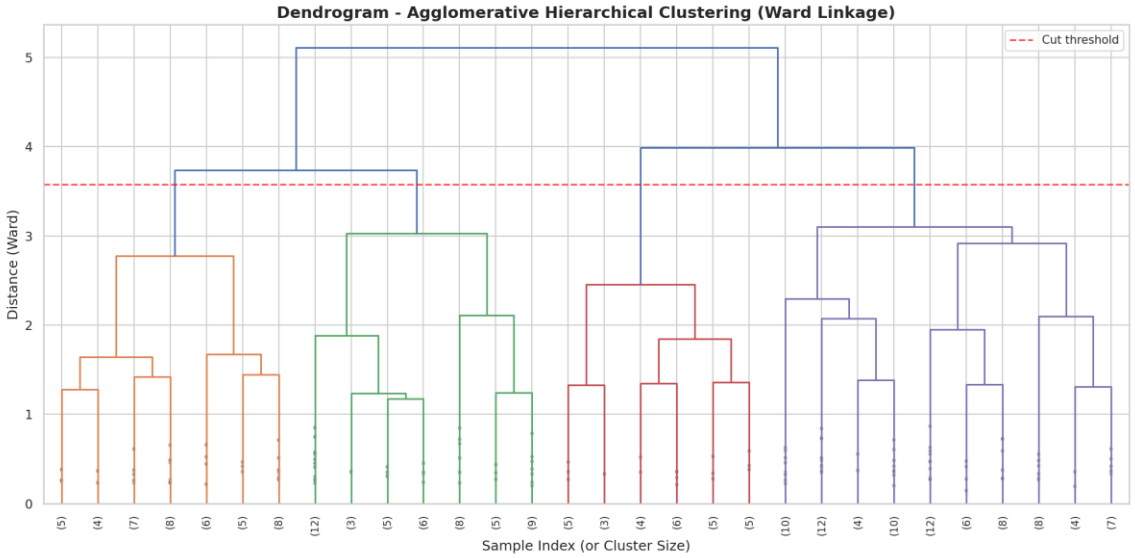


Figure 22: Dendrogram showing hierarchical clustering structure (Ward linkage, truncated).

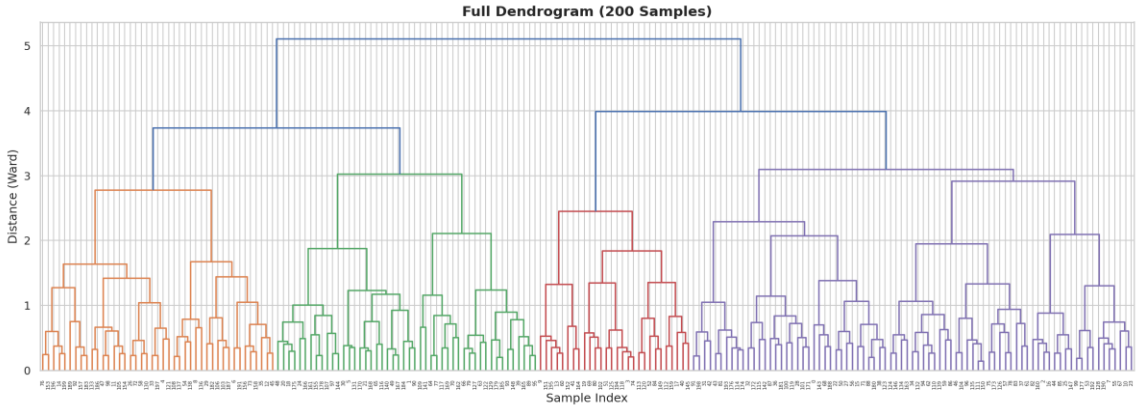


Figure 23: Full dendrogram showing all 200 samples.

With three clusters, the Agglomerative method produced cluster sizes of 91, 81, and 28 players. The silhouette score was 0.1122, which is very close to the KMeans k=3 score of 0.1144 without PCA. This tells us that both methods are finding a similar level of structure in the data. The third cluster is much smaller than the other two, which means the hierarchical method found a small group of players that are more different from the rest.

When comparing the Agglomerative clusters to the original EngagementLevel labels, the clusters do not line up very well. All three clusters contain a mix of High, Medium, and Low engagement players. This makes sense because the clustering was done on behavioral features (age, play time, sessions, level, achievements) without the label, and engagement level is probably influenced by things beyond just these numbers. The KMeans results showed a similar pattern. This is actually an interesting finding because it suggests that engagement level is not simply a function of the quantitative gaming behavior metrics alone.

Comparing all three clustering approaches, the KMeans with PCA (k=4) produced the best silhouette score at 0.2505. KMeans without PCA and Agglomerative both had similar, lower scores around 0.11-0.13. The main takeaway is that PCA helped the most, and the hierarchical approach gave us a different view of the data through the dendrogram, which shows how the merging happens step by step. Both methods agree that the data has some structure, but the clusters are not tightly defined. This is useful information for the supervised models coming up in later modules, because it suggests that predicting engagement level from these features alone may be challenging and additional features or different approaches may be needed.

Linear Regression Analysis

Linear regression was applied to Dataset 1 (Online Gaming Anxiety Study) to predict a player's GAD-7 anxiety score from a mix of gaming behavior and personality features. The GAD-7 score is a continuous numeric value between 0 and 21, which makes it a good fit for linear regression. The features used were Hours (weekly gaming hours), streams (hours per week watching gaming streams), Narcissism (single-item score), Age, SPIN_T (social phobia total), and SWL_T (satisfaction with life total). The goal was to see how well a simple linear model could predict anxiety from these inputs and which features matter the most.

Data Preparation

The cleaned Dataset 1 from the previous modules was used as the starting point. From the 43 columns in the cleaned dataset, six numeric features were chosen along with the GAD_T target. No further cleaning was needed because the missing values had already been handled in Module 1 with median imputation. The final dataset for linear regression has 11,233 rows and 7 columns (6 features plus the target). The data was split into a 75/25 train/test split, giving 8,424 training rows and 2,809 testing rows. A random_state of 42 was used so the split is reproducible.

Linear Regression Input Data (first 8 rows)

Hours	streams	Narcissism	Age	SPIN_T	SWL_T	GAD_T
15.0	0.0	1.0	25.0	5.0	23.0	1.0
8.0	2.0	1.0	41.0	33.0	16.0	8.0
0.0	0.0	4.0	32.0	31.0	17.0	8.0
20.0	5.0	2.0	28.0	11.0	17.0	0.0
20.0	1.0	1.0	19.0	13.0	14.0	14.0
4.0	0.0	2.0	24.0	13.0	17.0	1.0
30.0	8.0	2.0	29.0	26.0	16.0	0.0
2.0	0.0	1.0	23.0	55.0	12.0	12.0

Figure 25: Linear regression input data showing six features and the GAD_T target.

Model Results

The sklearn LinearRegression model was trained on the 6-feature input. The trained model produced an intercept of 5.998 and the following coefficients: Hours (+0.004), streams (+0.007), Narcissism (+0.315), Age (-0.014), SPIN_T (+0.125), and SWL_T (-0.182). The full linear equation that the model learned is shown below:

GAD_T = 5.998 + 0.004*Hours + 0.007*streams + 0.315*Narcissism – 0.014*Age + 0.125*SPIN_T – 0.182*SWL_T

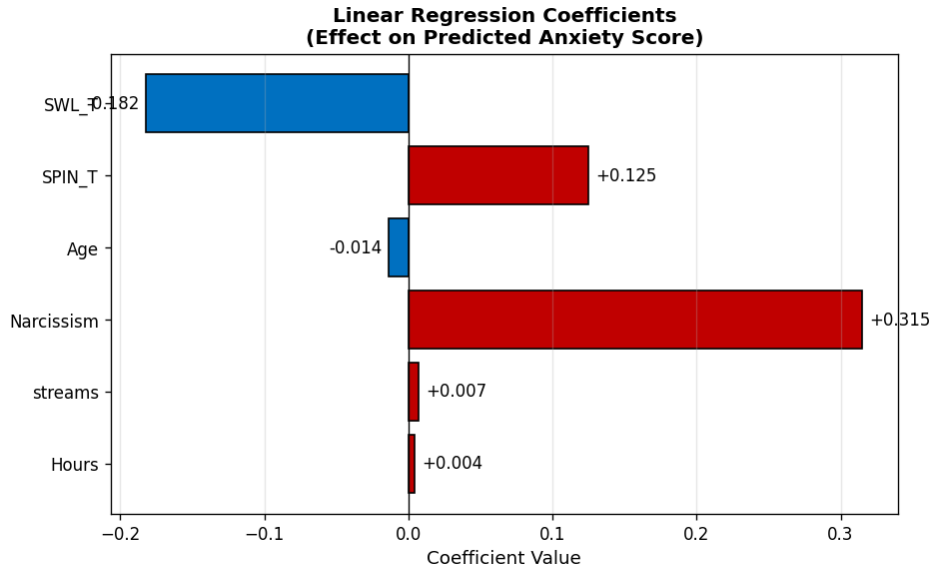


Figure 26: Coefficient values from the linear regression model. Larger bars mean a stronger effect on predicted anxiety. Positive bars push the prediction up, negative bars pull it down.

Worked Example

To show how the equation gets used, take an example player with these values: Hours=20, streams=5, Narcissism=3, Age=25, SPIN_T=25, SWL_T=20. Plugging those into the equation step by step:

Start with the intercept: 5.998

Add 0.004 * 20 = 0.082 → running total 6.079

Add 0.007 * 5 = 0.034 → running total 6.113

Add 0.315 * 3 = 0.945 → running total 7.058

Add –0.014 * 25 = –0.339 → running total 6.719

Add 0.125 * 25 = 3.115 → running total 9.834

Add –0.182 * 20 = –3.639 → final predicted GAD_T = 6.195

A predicted GAD-7 score of about 6.2 falls into the “Mild” anxiety range (5–9). The sklearn predict() function returned the same number (6.195), which confirms the math.

Results and Accuracy

The model was evaluated on the held-out test set of 2,809 players. The R-squared on the test set was 0.261, meaning the model explains about 26 percent of the variation in anxiety scores. RMSE on the test set was 4.03 GAD-7 points, and MAE was 3.07 GAD-

7 points. The training and test R-squared values were close (0.272 vs. 0.261), so the model is not overfitting. An R-squared of 0.26 is moderate. It is better than random guessing but it leaves a lot of room for improvement, which makes sense because anxiety is influenced by many things outside of gaming behavior. The actual vs predicted scatter plot below shows the predictions clustered around the diagonal line but with a wide spread, especially for higher anxiety scores.

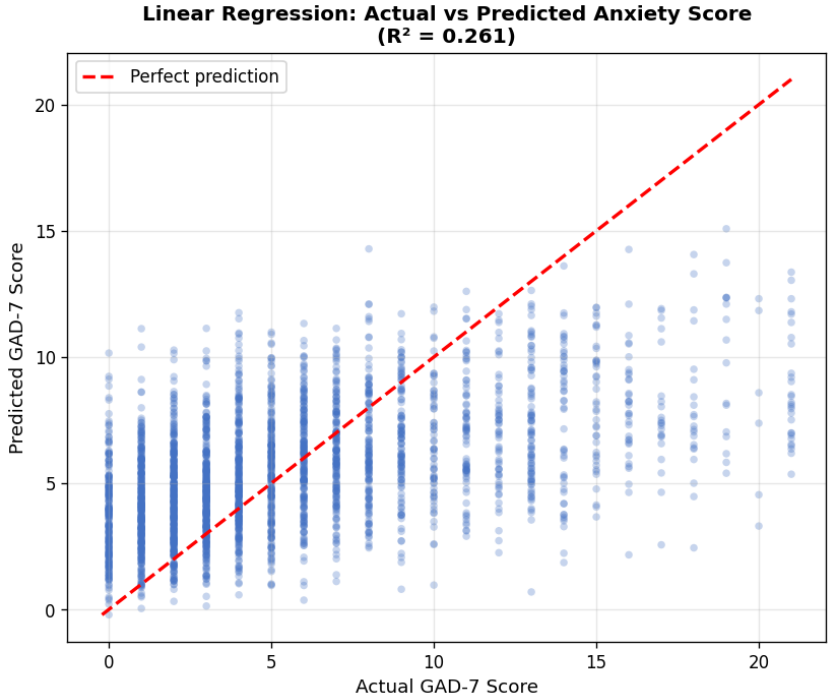


Figure 27: Actual vs predicted GAD-7 scores on the test set. The red dashed line is perfect prediction. Predictions track the true values but with notable scatter.

The residual plot below confirms the same story. Residuals (actual minus predicted) are mostly centered around zero, but the spread is wide and there is some structure where the model under-predicts the highest anxiety scores. This means the model captures the average trend but misses the extremes.

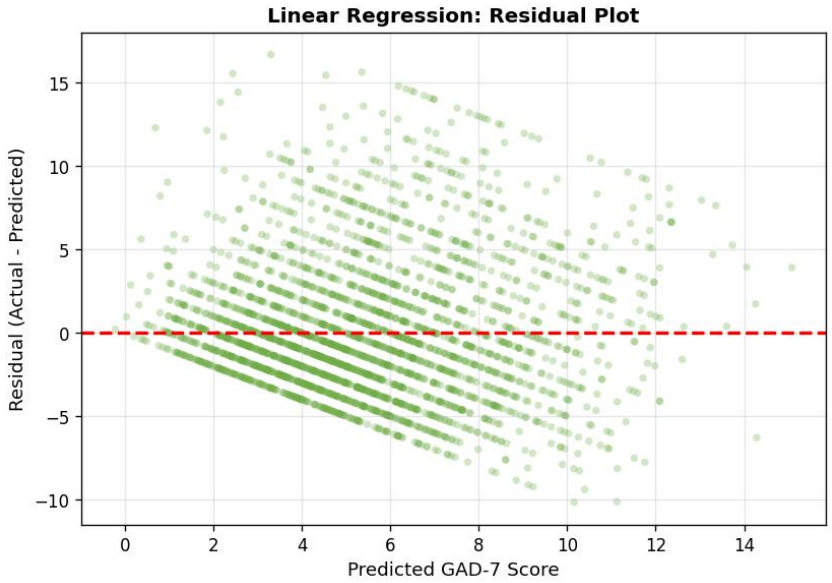


Figure 28: Residual plot showing prediction errors across the range of predicted scores. A flat band around zero means the model is unbiased, but the wide spread means single-player accuracy is limited.

Conclusions for Linear Regression

The linear regression results give a clear answer to one of the project's main questions: gaming hours alone is not a strong predictor of anxiety. The Hours coefficient came out to +0.004, which is essentially zero. Watching streams had a similarly tiny effect at +0.007. The features that mattered most were the other mental health and personality scores. Social phobia (SPIN_T) and life satisfaction (SWL_T) had the largest effects, with social phobia pushing anxiety up and life satisfaction pulling it down. Narcissism also pushed anxiety up by a smaller amount, and older age was tied to slightly lower anxiety. This lines up with what mental health research has been saying for a while, that anxiety is mostly about a person's overall mental and emotional state rather than how many hours they game. For Research Question 1, this is real evidence that there is no strong direct link between weekly gaming hours and GAD-7 scores. The takeaway for the project is that simply playing a lot of games does not mean a person is more anxious, and trying to use gaming behavior alone to predict anxiety is going to fall short. Mental health is shaped by many things, and the data backs that up.

Logistic Regression Analysis

Logistic regression was applied to Dataset 2 (Predict Online Gaming Behavior) to classify players as either “High” engagement or “Not High” engagement (Low or Medium combined). This problem is a natural fit for logistic regression because the target is a category rather than a continuous number. The features used were Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, AchievementsUnlocked, and InGamePurchases. The goal was to see how well behavioral signals can identify highly engaged players, and which behaviors matter the most.

Data Preparation

The 2,000-row sample of Dataset 2 from Module 1 was used as the input. The categorical EngagementLevel column was turned into a binary HighEngagement label, where High became 1 and both Medium and Low became 0. After this conversion, 509 players were labeled as High and 1,491 were labeled as Not High, giving a roughly 1-to-3 class ratio. All seven numeric features were standardized using StandardScaler so they all have a mean of 0 and a standard deviation of 1. This was important because the features were on very different scales (PlayTimeHours runs about 0–24 while PlayerLevel goes 1–99), and standardization makes the coefficients more comparable. A 75/25 train/test split was used with stratification on the label so both classes are represented in both sets.

Logistic Regression Input Data (first 8 rows)

Age	PlayTimeHours	SessionsPerWeek	AvgSessionDurationMinu	PlayerLevel	AchievementsUnlocke	InGamePurchases	HighEngagement
49.0	3.33	16.0	63.0	43.0	6.0	1.0	0.0
29.0	19.63	10.0	99.0	48.0	11.0	0.0	0.0
32.0	23.32	11.0	17.0	12.0	34.0	0.0	0.0
49.0	10.35	7.0	149.0	44.0	38.0	0.0	0.0
48.0	6.24	10.0	127.0	14.0	4.0	0.0	0.0
43.0	3.85	11.0	170.0	60.0	34.0	0.0	1.0
15.0	12.5	5.0	11.0	67.0	40.0	0.0	0.0
18.0	10.85	16.0	85.0	84.0	25.0	0.0	1.0

Figure 29: Logistic regression input data with seven features and the binary HighEngagement target (1 = High, 0 = Not High).

Model Results

The sklearn LogisticRegression model was trained on the standardized 7-feature input. The trained model produced an intercept of –2.376 and the following coefficients (on the standardized scale): Age (-0.026), PlayTimeHours (+0.013), SessionsPerWeek (+2.241), AvgSessionDurationMinutes (+2.023), PlayerLevel (+0.009),

AchievementsUnlocked (+0.103), and InGamePurchases (+0.097). The full linear part of the model is:

z = –2.376 – 0.026*Age + 0.013*PlayTimeHours + 2.241*SessionsPerWeek + 2.023*AvgSessionDurationMinutes + 0.009*PlayerLevel + 0.103*AchievementsUnlocked + 0.097*InGamePurchases

The probability of being in the High class is then computed by passing z through the sigmoid function:

P(High) = 1 / (1 + e^(-z))

If P(High) is 0.5 or higher the player is classified as High, otherwise as Not High. The coefficient chart below shows that two features dominate: SessionsPerWeek and AvgSessionDurationMinutes. The other five features have very small effects in comparison.

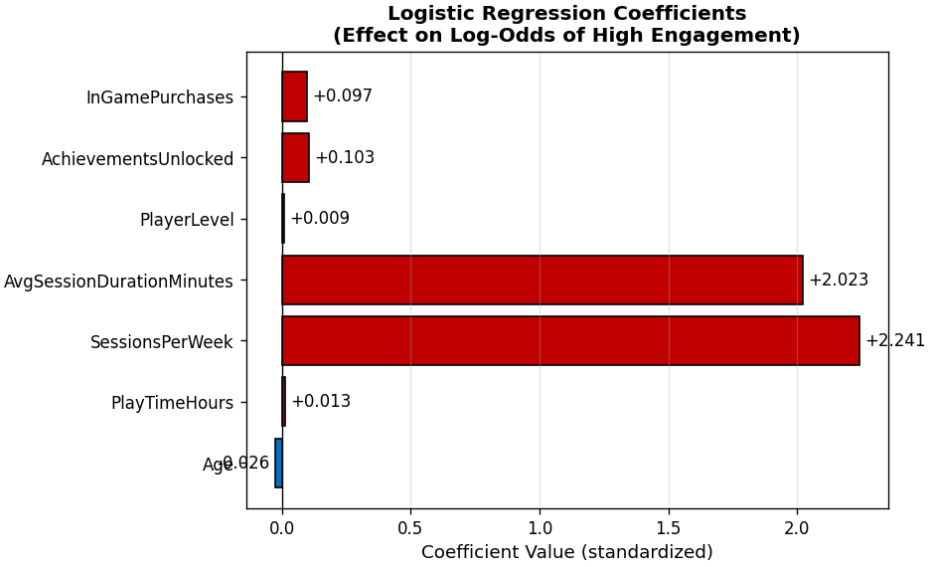


Figure 30: Logistic regression coefficients (standardized). SessionsPerWeek and AvgSessionDurationMinutes are by far the biggest drivers of high-engagement classification.

Worked Example

Take an example player with these raw values: Age=25, PlayTimeHours=15, SessionsPerWeek=18, AvgSessionDurationMinutes=90, PlayerLevel=75, AchievementsUnlocked=40, InGamePurchases=1. After standardization (using the scaler fit on the training data), these become approximately –0.706, 0.443, 1.497, –0.123, 0.892, 1.074, and 1.978. Plugging those into the linear part:

Start: –2.376

Add (–0.026)(–0.706) = +0.019 → z = –2.357

Add (+0.013)(0.443) = +0.006 → z = –2.352

Add (+2.241)(1.497) = +3.355 → z = +1.004

Add (+2.023)(–0.123) = –0.248 → z = +0.755

Add (+0.009)(0.892) = +0.008 → z = +0.763

Add (+0.103)(1.074) = +0.111 → z = +0.874

Add (+0.097)(1.978) = +0.193 → final z = +1.067

Now apply the sigmoid: P(High) = 1 / (1 + e^(–1.067)) = 1 / (1 + 0.344) = 0.744. Because 0.744 is greater than the 0.5 threshold, the model classifies this player as High engagement. The sklearn predict_proba() function returned the same probability of 0.744, which confirms the calculation.

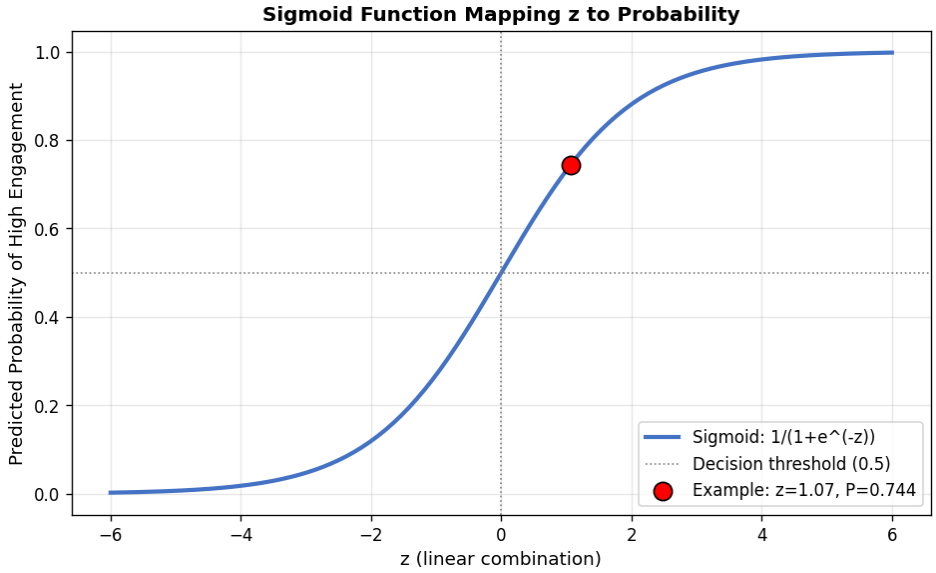


Figure 31: Sigmoid curve mapping z to probability. The red dot marks the worked example at z = 1.07 and P = 0.744, which falls above the 0.5 threshold and is classified as High engagement.

Results and Accuracy

The model was evaluated on the held-out test set of 500 players. The overall accuracy was 0.932, meaning the model correctly classified about 93 percent of players. The confusion matrix shows that out of the 373 actual Not-High players, the model got 361 right and only 12 wrong. Out of the 127 actual High players, it got 105 right and missed 22. Precision for the High class was 0.897, recall was 0.827, and F1 score was 0.861. These numbers are strong, especially for a basic logistic regression with no tuning. The

minority class (High) is the harder one to predict, and 22 missed High players means the recall is the main weak spot.

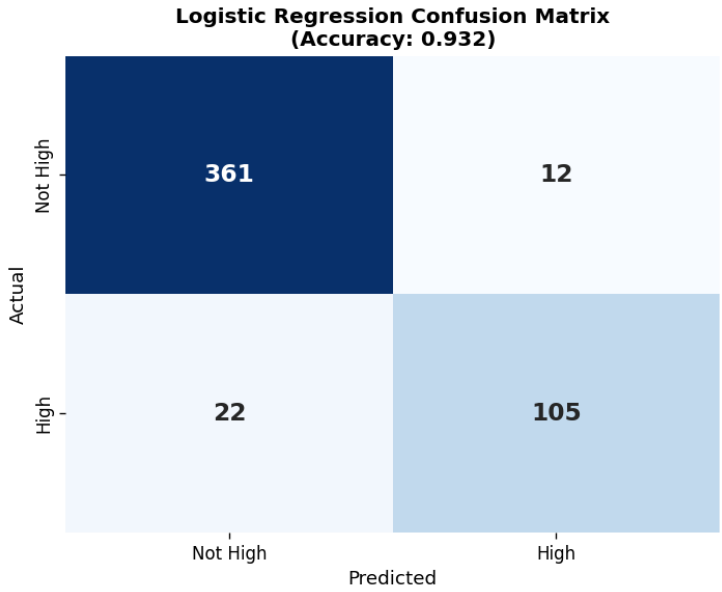


Figure 32: Confusion matrix for the logistic regression model on the test set. Most errors come from missing actual High-engagement players (false negatives in the bottom-left).

The ROC curve below adds more context by showing how the true positive rate trades off against the false positive rate as the decision threshold changes. The area under the curve (AUC) was 0.935, which is well above the 0.5 baseline of a random classifier and indicates strong discrimination between the two classes.

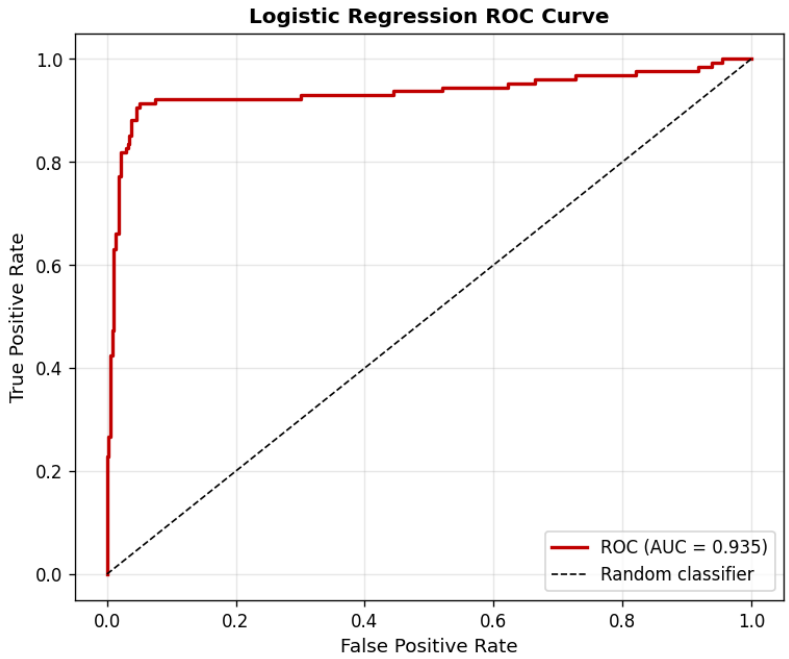


Figure 33: ROC curve for the logistic regression model. AUC = 0.935 indicates strong separation between High and Not-High classes.

Conclusions for Logistic Regression

The logistic regression results show that high engagement can be predicted with strong accuracy when the right behavioral features are available. Two features did almost all the heavy lifting: SessionsPerWeek and AvgSessionDurationMinutes. Together these two coefficients dwarfed the rest, which lines up with intuition. A player who logs in often and stays for long sessions is more engaged than someone who plays rarely or briefly, no matter how many achievements they have or how much they spend. Total PlayTimeHours barely mattered on its own, which is interesting because it suggests that pattern matters more than total volume. Two short sessions a week of three hours each is very different from twelve thirty-minute sessions, even though the total play time is similar. For Research Question 10, this gives a concrete answer: session frequency and session length are the best predictors of engagement among the features tested. The 93 percent accuracy and 0.935 AUC also suggest that engagement, unlike anxiety, is much more predictable from behavioral data alone. This contrast between the linear regression result and the logistic regression result is one of the most important takeaways from this module.

Decision Tree Classification

Decision tree classification was applied to Dataset 2 (Predict Online Gaming Behavior) to classify players into one of three engagement levels: Low, Medium, or High. Decision trees are a natural fit for this kind of problem because they handle both numeric and categorical features, they produce results that are easy to follow visually, and they work well on multi-class targets without needing any special setup. The goal here is to compare two different tree configurations and see how tree depth and split criterion affect both accuracy and model complexity.

Data Preparation

The 2,000-row sample of Dataset 2 from Module 1 was used as the input. All eleven features were included: the seven numeric ones (Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, AchievementsUnlocked, InGamePurchases) and the four categorical ones (Gender, Location, GameGenre, GameDifficulty). The categorical columns were ordinal-encoded into integer values using sklearn's OrdinalEncoder so the tree could split on them. For example, the GameGenre column has five values that were encoded as: Action=0, RPG=1, Simulation=2, Sports=3, Strategy=4. The target was the three-class EngagementLevel column (Low, Medium, High). A 75/25 train/test split was used with stratification on the label so all three classes are represented in both sets. This gave 1,500 training rows and 500 testing rows.

Decision Tree Input Data (first 8 rows, categoricals ordinal-encoded)

Age	PlayTimeHours	SessionsPerWeek	AvgSessionDurationMi	PlayerLevel	AchievementsUnlock	InGamePurchases	Gender	Location	GameGenre	GameDifficulty	EngagementLevel
49	3.33	16	63	43	6	1	1.0	2.0	0.0	2.0	Medium
29	19.63	10	99	48	11	0	0.0	3.0	0.0	1.0	Medium
32	23.32	11	17	12	34	0	1.0	0.0	3.0	2.0	Medium
49	10.35	7	149	44	38	0	1.0	0.0	0.0	2.0	Medium
48	6.24	10	127	14	4	0	0.0	2.0	2.0	0.0	Medium
43	3.85	11	170	60	34	0	0.0	0.0	4.0	0.0	High
15	12.5	5	11	67	40	0	1.0	0.0	0.0	2.0	Low
18	10.85	16	85	84	25	0	1.0	1.0	2.0	1.0	High

Figure 34: Decision tree input data showing all 11 features and the EngagementLevel target. The categorical columns appear as numeric codes after ordinal encoding.

Model Results

Two different decision trees were trained with different parameters to compare a simple, shallow model against a deeper, more complex one. Tree 1 used max_depth=3 with the Gini impurity criterion. Tree 2 used max_depth=7 with the Entropy criterion. Both used the same random_state of 42 for reproducibility.

Tree 1: Shallow Gini Tree

Tree 1 reached a depth of 3 with 8 leaves. The root node split on SessionsPerWeek at a threshold of 2.5, and the two features that did almost all the work were SessionsPerWeek and AvgSessionDurationMinutes. These two together account for over 95 percent of the feature importance in this tree. The visualization below shows the full structure with the Gini score, sample count, and class distribution at each node.

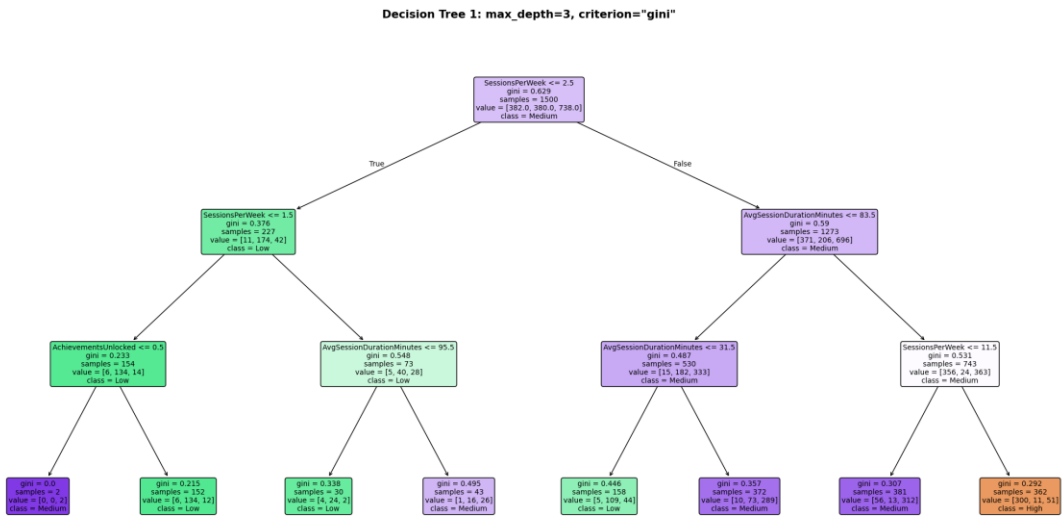


Figure 35: Decision Tree 1 (max_depth=3, criterion="gini"). Color intensity in each node shows which class dominates: green for High, blue for Low, orange for Medium.

Tree 1 Worked Example

To show how a player gets classified, take test sample 0 with these key values: SessionsPerWeek=18, AvgSessionDurationMinutes=127, PlayerLevel=25. The actual class for this player is High. Tracing through Tree 1 step by step:

Root: Is SessionsPerWeek ≤ 2.5? 18 is greater than 2.5, so go RIGHT.

Next: Is AvgSessionDurationMinutes ≤ 83.5? 127 is greater than 83.5, so go RIGHT.

Next: Is SessionsPerWeek ≤ 11.5? 18 is greater than 11.5, so go RIGHT.

Leaf reached. The class distribution at this leaf is roughly 82.9 percent High, 14.1 percent Medium, and 3.0 percent Low. The tree predicts High, which matches the actual class. This shows how just two questions (about session frequency and session length) can already get most of the way to a correct High-engagement prediction.

Tree 1 Confusion Matrix and Accuracy

Tree 1 was evaluated on the held-out test set of 500 players. Overall accuracy was 0.746, meaning it got about 75 percent of the predictions right. Looking at the confusion

matrix, the tree did best on the Medium class (the majority class) and was a bit weaker on Low. Out of 127 actual Low-engagement players, it caught 78 correctly. Out of 127 actual High-engagement players, it got 93 correct. Macro F1 was around 0.74.

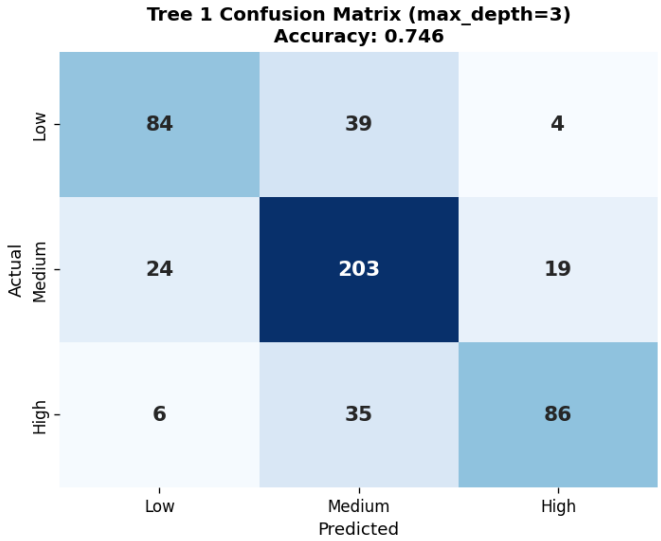


Figure 36: Confusion matrix for Tree 1. The diagonal shows correct predictions for each class.

Tree 2: Deeper Entropy Tree

Tree 2 was trained with max_depth=7 and the Entropy criterion to see if a deeper, more flexible tree could do better. It ended up with a depth of 7 and 60 leaves, which is a lot more than Tree 1. The same two features (SessionsPerWeek and AvgSessionDurationMinutes) still dominated, but several other features (PlayerLevel, AchievementsUnlocked, PlayTimeHours) started to contribute small amounts of importance. The diagram below shows the top four levels of Tree 2 for readability since the full tree is too dense to display cleanly.

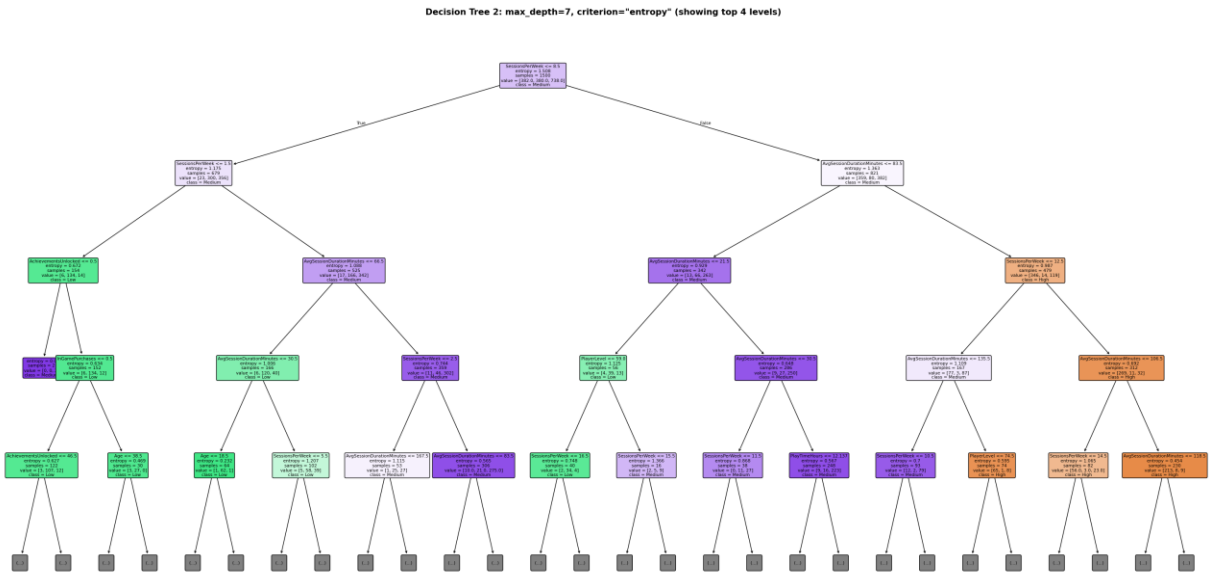


Figure 37: Decision Tree 2 (max_depth=7, criterion="entropy", top 4 levels shown). The tree branches out into many more leaves at lower levels.

Tree 2 Confusion Matrix and Accuracy

Tree 2 reached an accuracy of 0.824 on the test set, about 8 percentage points higher than Tree 1. The deeper structure let it pick up on more subtle patterns, especially in the gray zone between Low and Medium engagement. The confusion matrix shows clear improvement across all three classes. The main risk with a deeper tree is overfitting, but here the gap between training and test accuracy was small enough that depth 7 looks like a reasonable trade-off.

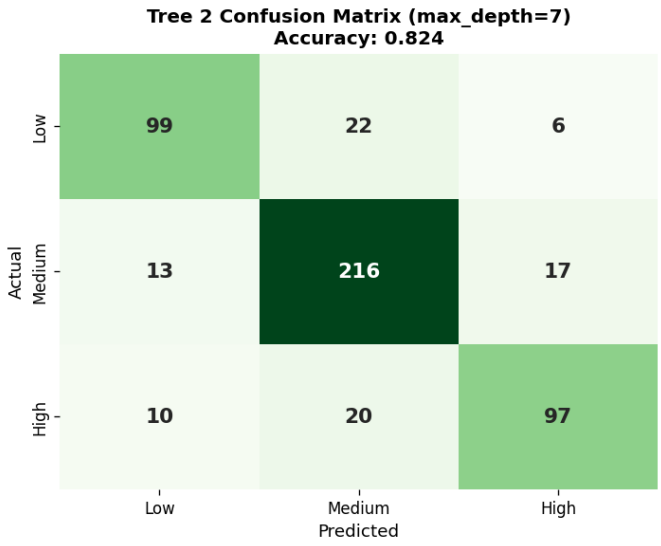


Figure 38: Confusion matrix for Tree 2. Accuracy is noticeably higher than Tree 1, especially for the Low and High classes.

Feature Importance Comparison

Both trees agreed strongly on which features matter most. SessionsPerWeek and AvgSessionDurationMinutes together account for over 90 percent of the feature importance in both trees. This lines up exactly with what the logistic regression model showed in the previous module. The categorical features (Gender, Location, GameGenre, GameDifficulty) had almost zero importance in either tree, which suggests they don't carry meaningful information about engagement level.

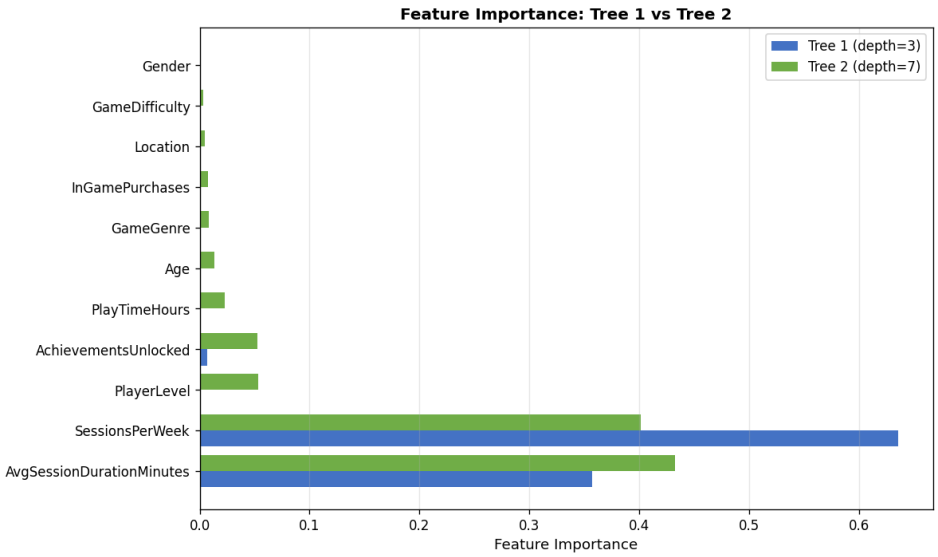


Figure 39: Feature importance comparison between Tree 1 and Tree 2. Both trees agree that session frequency and session length are by far the most useful features.

Conclusions for Decision Trees

The decision tree results back up what the earlier models already suggested: engagement level is driven mainly by how often a player logs in and how long they stay. The shallow Tree 1 captured the main pattern with just three splits and hit 75 percent accuracy. The deeper Tree 2 picked up extra patterns and pushed accuracy up to 82 percent, though at the cost of being much harder to read. For Research Question 10 on which features best predict engagement, the trees give the same answer as the logistic regression: SessionsPerWeek and AvgSessionDurationMinutes are the heavy hitters. One nice thing about decision trees is that they make the rules explicit. The path through Tree 1 reads almost like a simple checklist a human could follow: if a player has lots of sessions per week and long sessions, predict High. That kind of clarity is valuable for explaining the model to a non-technical audience, even when the accuracy is slightly lower than a more complex model.

Bayesian Classification

Two Naive Bayes classifiers were applied to Dataset 2 to predict EngagementLevel: Gaussian Naive Bayes on the numeric features, and Categorical Naive Bayes on the categorical and discretized features. These two methods were chosen because they use very different assumptions about the data. Gaussian NB assumes each numeric feature is normally distributed within each class. Categorical NB assumes each feature takes one of a small set of discrete values and learns probabilities for each value within each class. Running both lets us compare how the same dataset performs under each method.

Method 1: Gaussian Naive Bayes

Gaussian Naive Bayes works on continuous numeric data. For each class, the model learns the mean and variance of each feature, then assumes the feature values follow a normal (Gaussian) distribution around that mean. To classify a new player, it calculates how likely the observed feature values are under each class and combines that with the prior probability of each class using Bayes' theorem. The class with the highest posterior probability wins.

Data Preparation for Gaussian NB

The seven numeric features from Dataset 2 were used as input: Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, AchievementsUnlocked, and InGamePurchases. No scaling or normalization was needed because Naive Bayes is not sensitive to feature scales. A 75/25 train/test split was used with stratification on the EngagementLevel label.

GaussianNB Input Data (numeric features, first 8 rows)

Age	PlayTimeHours	SessionsPerWeek	AvgSessionDurationMinu	PlayerLevel	AchievementsUnlocked	InGamePurchases	EngagementLevel
49	3.33	16	63	43	6	1	Medium
29	19.63	10	99	48	11	0	Medium
32	23.32	11	17	12	34	0	Medium
49	10.35	7	149	44	38	0	Medium
48	6.24	10	127	14	4	0	Medium
43	3.85	11	170	60	34	0	High
15	12.5	5	11	67	40	0	Low
18	10.85	16	85	84	25	0	High

Figure 40: GaussianNB input data (numeric features only, first 8 rows). All values remain in their original units.

Model Results and Probabilities

The model learned class priors of P(Low)=0.253, P(Medium)=0.492, and P(High)=0.255. These match the natural class distribution in the data. For each class,

the model also learned a mean for every feature. The most telling differences were in SessionsPerWeek (mean of 14.2 for High vs 4.8 for Low) and AvgSessionDurationMinutes (mean of 133.8 for High vs 66.9 for Low). These two features have very different distributions across classes, which is what gives the model its predictive power.

To show how a prediction works, take test sample 0 with these values: Age=47, PlayTimeHours=10.55, SessionsPerWeek=18, AvgSessionDurationMinutes=127, PlayerLevel=25, AchievementsUnlocked=39, InGamePurchases=1. The actual class is High. The model computes a likelihood (using the Gaussian PDF) for each feature under each class, multiplies them together with the class prior, and normalizes to get the posterior probabilities. The result was P(High|x)=0.787, P(Low|x)=0.008, P(Medium|x)=0.205. The High probability is by far the highest, so the model predicts High, which is correct. The 0.787 probability means the model is fairly confident, though not totally certain. A 0.987 would mean near-certainty; a 0.50 would mean a coin flip.

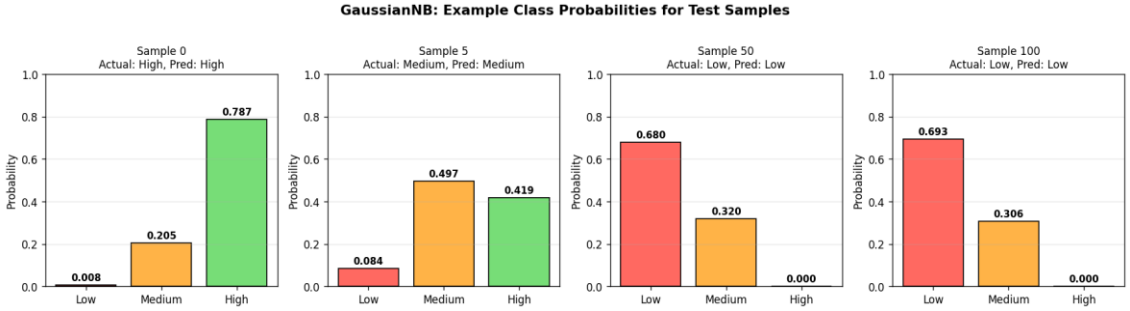


Figure 41: GaussianNB class probabilities for four example test samples. Each bar set shows P(Low), P(Medium), and P(High). The model is most confident when one bar dominates.

Gaussian NB Accuracy

GaussianNB achieved an overall accuracy of 0.800 on the test set, which is right in between Tree 1 (0.746) and Tree 2 (0.824). The confusion matrix shows it did very well on the Medium class (238 of 246 correct) but had some trouble with Low (73 of 127 correct, with 51 misclassified as Medium). This pattern suggests that Low-engagement players and Medium-engagement players have somewhat overlapping numeric profiles, which makes them harder to tell apart. The High class did well at 89 of 127 correct.

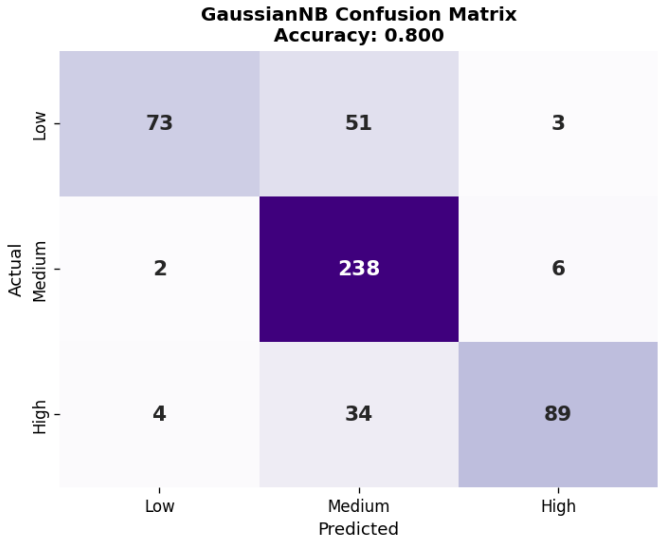


Figure 42: Confusion matrix for Gaussian Naive Bayes. The strongest performance is on the Medium class; Low is the weakest because many Low players get classified as Medium.

Method 2: Categorical Naive Bayes

Categorical Naive Bayes is designed for inputs where every feature takes a discrete value from a small set, like a category code. Instead of assuming a Gaussian distribution, it learns, for each class and each feature, the probability of seeing each possible category. To make this work with the gaming dataset, the four natural categorical features were combined with three discretized numeric features, turning continuous values into bins.

Data Preparation for Categorical NB

Seven categorical inputs were used. Four came directly from the raw dataset: Gender, Location, GameGenre, and GameDifficulty. Three more were created by discretizing the most important numeric features into bins. SessionsPerWeek was binned into VeryLow (0-5), Low (6-10), Med (11-15), High (16-25). AvgSessionDurationMinutes was binned into Short (0-45), Med (46-90), Long (91-135), VeryLong (136-200). PlayerLevel was binned into Beginner (0-25), Inter (26-50), Advanced (51-75), Expert (76-100). All seven features were then ordinal-encoded into integers, which is the format CategoricalNB expects.

CategoricalNB Input Data (raw categories + discretized bins, first 8 rows)

Gender	Location	GameGenre	GameDifficulty	SessionsPerWeek_bin	AvgSession_bin	PlayerLevel_bin	EngagementLevel
Male	Other	Action	Medium	High	Med	Inter	Medium
Female	USA	Action	Hard	Low	Long	Inter	Medium
Male	Asia	Sports	Medium	Med	Short	Beginner	Medium
Male	Asia	Action	Medium	Low	VeryLong	Inter	Medium
Female	Other	Simulation	Easy	Low	Long	Beginner	Medium
Female	Asia	Strategy	Easy	Med	VeryLong	Advanced	High
Male	Asia	Action	Medium	VeryLow	Short	Advanced	Low
Male	Europe	Simulation	Hard	High	Med	Expert	High

Figure 43: CategoricalNB input data before encoding, showing the raw categorical values and the discretized bins (first 8 rows).

CategoricalNB Input Data (ordinal-encoded integers, first 8 rows)

Gender	Location	GameGenre	GameDifficulty	SessionsPerWeek_bin	AvgSession_bin	PlayerLevel_bin	EngagementLevel
1	2	0	2	0	1	3	Medium
0	3	0	1	1	0	3	Medium
1	0	3	2	2	2	1	Medium
1	0	0	2	1	3	3	Medium
0	2	2	0	1	0	1	Medium
0	0	4	0	2	3	0	High
1	0	0	2	3	2	0	Low
1	1	2	1	0	1	2	High

Figure 44: Same data after ordinal encoding into integer codes. CategoricalNB needs integer-coded categorical inputs.

Model Results and Probabilities

The trained CategoricalNB model used the same class priors as GaussianNB (P(Low)=0.253, P(Medium)=0.492, P(High)=0.255). For each feature, it learned a probability table that gives the chance of seeing each category given the class. For instance, in the High class, the SessionsPerWeek bin “High (16-25)” is far more common than it is in the Low class, where the “VeryLow (0-5)” bin dominates. Bayes’ theorem then combines all of these category-given-class probabilities with the class priors to produce posterior probabilities for a new player.

For the same test sample 0 as before, CategoricalNB produced P(High|x)=0.644, P(Low|x)=0.037, P(Medium|x)=0.320. It also predicts High, which matches the actual class. Notice the probability for High is lower (0.644 vs 0.787) and the probability for Medium is higher (0.320 vs 0.205) compared to GaussianNB. This is because the discretization loses some of the fine-grained information that GaussianNB could pick up on. Another test sample (sample 5) shows what an uncertain prediction looks like: P(High|x)=0.468, P(Medium|x)=0.461. The two are almost tied, so the model picks High because it has a slight edge, but the truth is actually Medium. Probabilities this close are a sign of a borderline case.

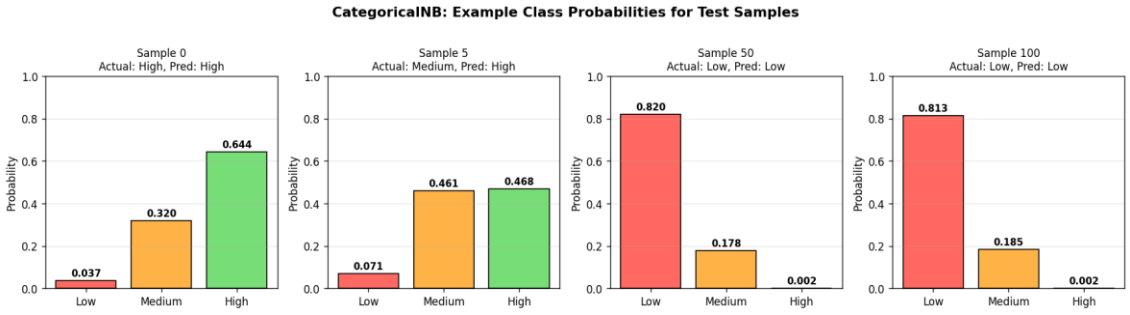


Figure 45: CategoricalNB class probabilities for four example test samples. When two bars are close to equal height (like sample 5), the model is not very sure of its answer.

Categorical NB Accuracy

CategoricalNB reached an accuracy of 0.698 on the test set, which is the lowest of the four classifiers tried so far. It did well on Medium (210 of 246) and reasonably on High (86 of 127), but struggled with Low (53 of 127). The drop in accuracy compared to GaussianNB is a clear cost of discretizing the numeric features into bins. The bin boundaries throw away information about where exactly a player's SessionsPerWeek value falls within the bin, and that information turned out to be useful.

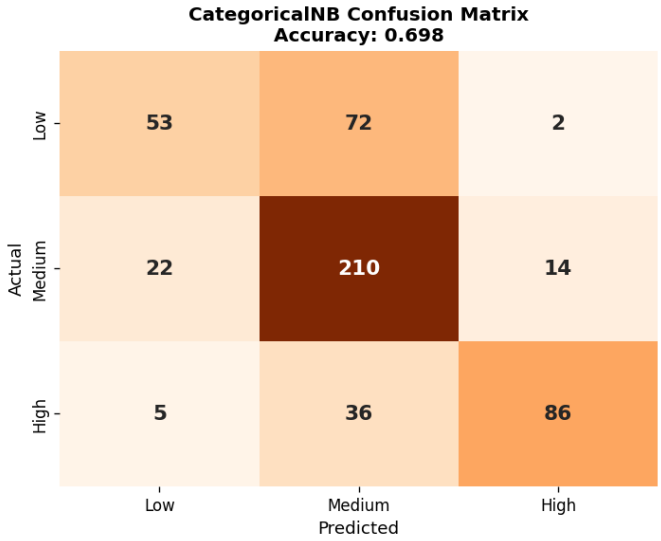


Figure 46: Confusion matrix for Categorical Naive Bayes. The model loses accuracy compared to GaussianNB, especially on the Low class.

Conclusions for Naive Bayes

The two Naive Bayes methods told a consistent story. Both correctly identified that session frequency and session length are the most informative signals for engagement. GaussianNB did meaningfully better than CategoricalNB (80 percent vs 70 percent accuracy), which makes sense because the numeric features carry continuous information that gets lost when you bin them. The probability outputs from both models are useful for understanding how confident the model is on any given prediction. A probability of 0.80 means the model is fairly sure; a probability of 0.45 means it is basically guessing between two classes. This is one of the strengths of Naive Bayes over a basic decision tree: it doesn't just give you a class label, it gives you a number that shows how strong the evidence is. For the project topic, both Naive Bayes models support the conclusion that engagement level is largely predictable from a few key behavioral patterns, and that adding raw demographic and game-genre categories does not help much.

Support Vector Machine Classification

Support Vector Machine (SVM) classification was applied to Dataset 2 (Predict Online Gaming Behavior) to classify players into the three engagement levels: Low, Medium, and High. SVMs work by finding the boundary (or boundaries) that best separate the classes while leaving the widest possible margin between them. They require labeled, quantitative data, which fits this task well since all seven numeric features are quantitative and the EngagementLevel label is known. Three different kernels were tested (linear, RBF, and polynomial) to see which one best captures the shape of the data.

Data Preparation

The seven numeric features from Dataset 2 were used: Age, PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, PlayerLevel, AchievementsUnlocked, and InGamePurchases. Unlike the tree-based models, SVMs are sensitive to feature scale because they rely on distances between points. For that reason, all seven features were standardized using StandardScaler so each has a mean of 0 and a standard deviation of 1. Without this step, a large-range feature like PlayerLevel (1 to 99) would overwhelm a small-range feature like InGamePurchases (0 or 1). A 75/25 train/test split with stratification gave 1,500 training rows and 500 testing rows.

Standardized Input Data for SVM (first 8 rows)

Age	PlayTimeHours	SessionsPerWeek	AvgSessionDurationMinu	PlayerLevel	AchievementsUnlocked	InGamePurchases	EngagementLevel
-0.406	0.817	-1.135	0.385	-0.047	1.695	-0.505	Medium
0.396	1.164	1.497	0.202	0.962	0.453	1.978	High
1.498	1.601	-0.082	-0.306	-0.117	-0.514	-0.505	Medium
0.296	1.206	-1.311	0.629	-0.883	0.798	-0.505	Low
1.598	1.425	1.146	-0.265	0.683	0.384	-0.505	High
-0.906	0.013	-0.082	-0.041	-0.569	1.005	-0.505	Medium
-0.005	0.483	0.093	0.243	0.022	-0.376	1.978	Medium
0.797	0.871	-1.311	1.543	-1.37	-0.031	-0.505	Medium

Figure 47: Standardized input data for the SVM models (first 8 rows). Every feature is now on the same scale, centered around 0.

Model Results: Three Kernels

Three SVM models were trained, one for each kernel, all using a regularization cost of C=1.0. The cost parameter C controls the trade-off between a wide margin and correctly classifying the training points. A small C allows a wider margin with more mistakes; a large C forces the model to classify more training points correctly at the risk of overfitting. C=1.0 is a balanced default. The kernel determines the shape of the decision boundary: the linear kernel draws straight separating lines, the RBF (radial basis function) kernel can wrap around clusters with curved boundaries, and the polynomial kernel uses curved boundaries based on a degree-3 polynomial.

Linear Kernel

The linear kernel reached an accuracy of 0.782. It tries to separate the three engagement classes using straight-line boundaries in the seven-dimensional feature space. The confusion matrix below shows it did reasonably well but had the most trouble of the three kernels, mostly mixing up Low and Medium players. This suggests the true boundaries between classes are not perfectly straight.

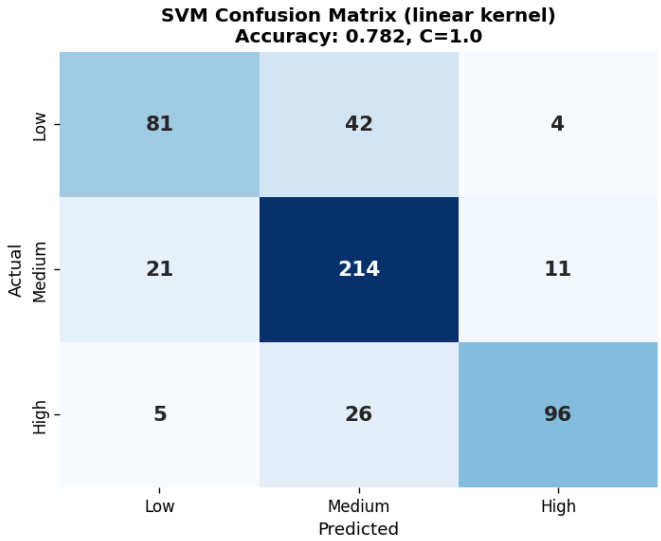


Figure 48: Confusion matrix for the linear-kernel SVM (C=1.0, accuracy 0.782).

RBF Kernel

The RBF kernel was the best performer of the three, with an accuracy of 0.836. The RBF kernel can form curved, flexible boundaries that wrap around groups of points, which lets it capture more complex separation between the classes. The confusion matrix shows clear improvement over the linear kernel, especially on the Low and High classes. This tells us the engagement classes are separated by curved rather than straight boundaries.

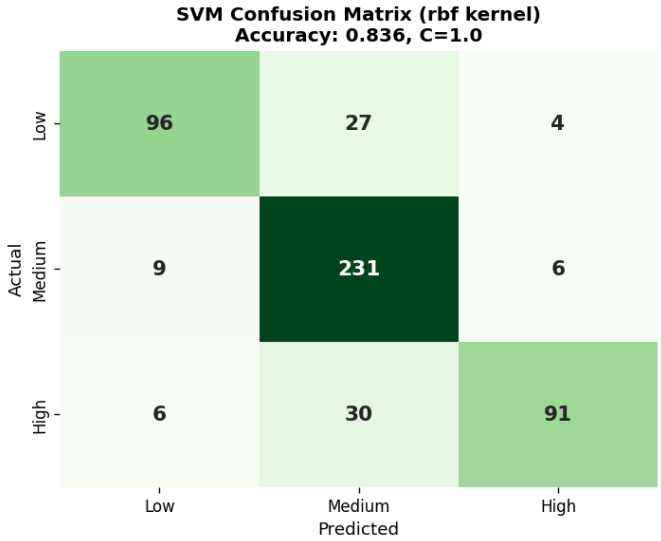


Figure 49: Confusion matrix for the RBF-kernel SVM (C=1.0, accuracy 0.836). This was the strongest of the three kernels.

Polynomial Kernel

The polynomial kernel (degree 3) reached an accuracy of 0.792, landing between the linear and RBF kernels. It uses curved boundaries based on polynomial combinations of the features. While it beat the linear kernel, it did not match the RBF kernel on this data. The confusion matrix shows a pattern similar to the linear kernel, with most errors again coming from the Low and Medium overlap.

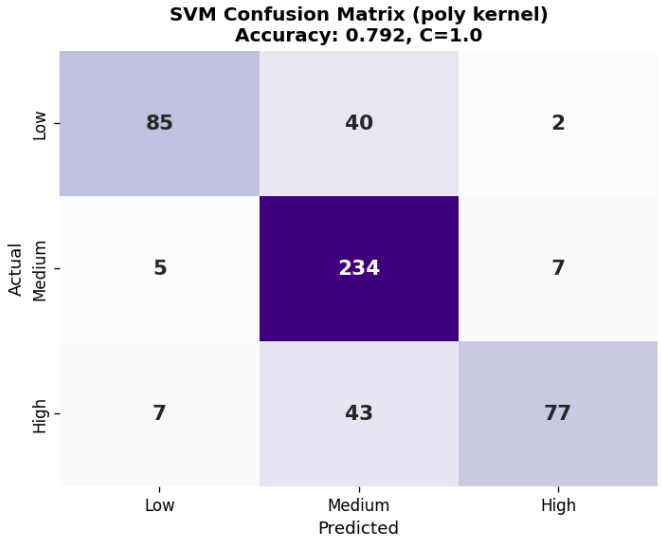


Figure 50: Confusion matrix for the polynomial-kernel SVM (degree 3, C=1.0, accuracy 0.792).

Kernel Comparison

Comparing the three kernels side by side, all used the same cost of C=1.0, so the only difference was the shape of the decision boundary. The RBF kernel won at 0.836, followed by polynomial at 0.792 and linear at 0.782. The clear gap between RBF and the other two shows that the engagement classes in this dataset are best separated by flexible, curved boundaries rather than straight lines or simple polynomial curves. The chart below makes the comparison easy to see at a glance.

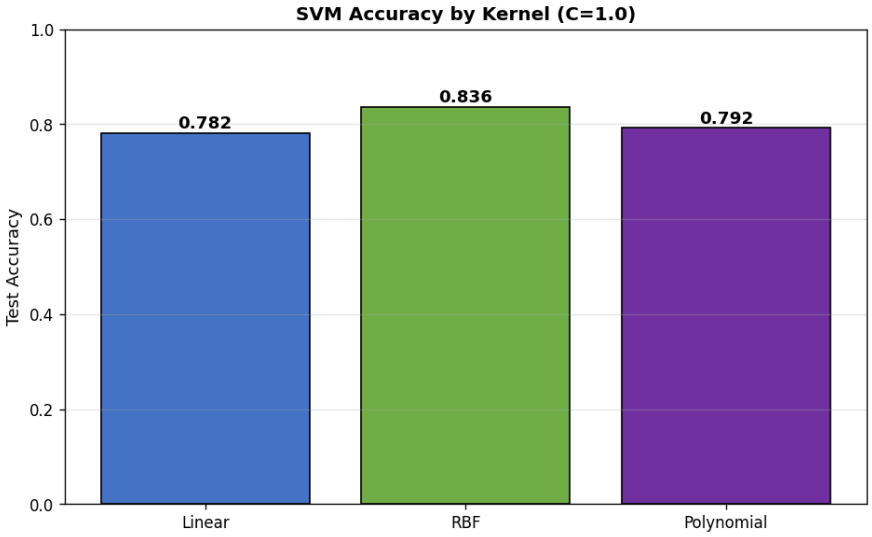


Figure 51: SVM accuracy comparison across the three kernels (all at C=1.0). The RBF kernel clearly outperformed the others.

Conclusions for SVM

The SVM results add another angle to the project's engagement question. All three kernels were able to classify players reasonably well, which once again confirms that engagement level is predictable from behavioral data. The fact that the RBF kernel did noticeably better than the linear one is the interesting part. It means the line between a Low, Medium, and High engagement player is not a simple straight cut. Instead, the groups blend into each other in a curved, more complicated way, and a flexible model is needed to trace that boundary. In plain terms, you cannot just draw a single straight line and say "everyone above this many sessions per week is highly engaged." The reality is fuzzier than that, and the RBF kernel handles the fuzziness best. This is consistent with the clustering results from earlier in the project, which also showed that the player groups overlap rather than forming clean, separate blobs.

Ensemble Method Classification

Ensemble methods combine many models into one stronger model. Instead of relying on a single decision tree or a single classifier, ensembles pool the predictions of many models to get better accuracy and more stable results. Three ensemble methods were applied to Dataset 2 to classify EngagementLevel: Random Forest, AdaBoost, and Stacking. All three used the same seven numeric features and the same 75/25 train/test split as the earlier models, so their results can be compared directly.

Random Forest

Random Forest builds many decision trees, each trained on a random subset of the data and features, and then has them vote on the final prediction. This “wisdom of the crowd” approach usually beats a single tree because the errors of individual trees tend to cancel out. The model here used 100 trees with a maximum depth of 10 each.

Random Forest Data Preparation

The same seven numeric features were used. Random Forest does not require scaling because it splits on one feature at a time and is not affected by feature ranges, so the raw (unscaled) values were used. The 75/25 stratified split gave 1,500 training rows and 500 testing rows.

Random Forest Results

The Random Forest reached an accuracy of 0.850, the best of any single model in the project so far. The confusion matrix shows strong performance across all three classes, with the Medium class predicted especially well. The model’s feature importance scores once again put SessionsPerWeek (0.372) and AvgSessionDurationMinutes (0.327) at the top, matching every other model in this project. The remaining five features each contributed under 10 percent.

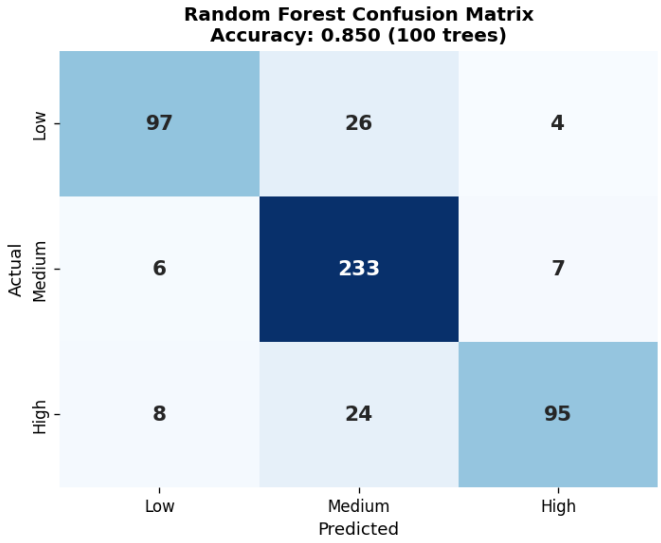


Figure 52: Confusion matrix for the Random Forest (100 trees, max_depth=10, accuracy 0.850).

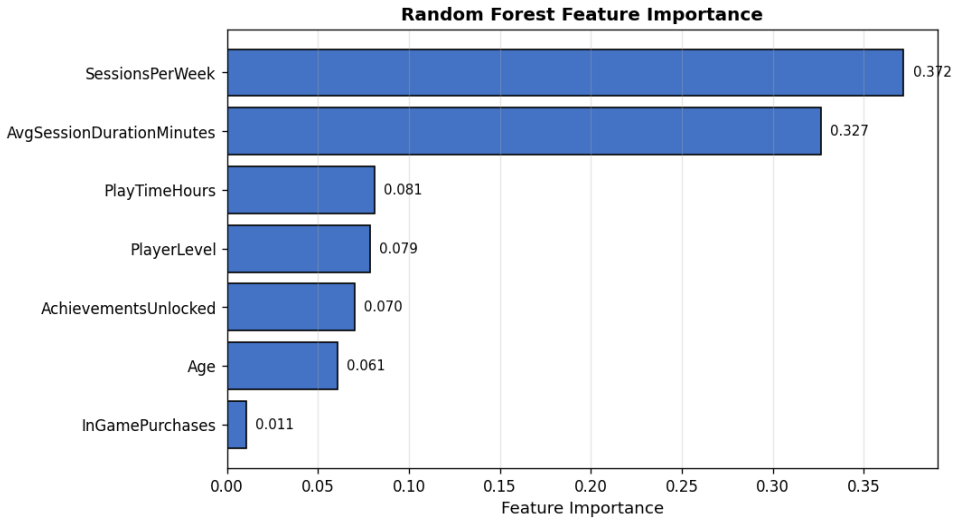


Figure 53: Random Forest feature importance. Session frequency and session length once again dominate.

Visualizing Three Trees

A full Random Forest with 100 trees is impossible to display, so a smaller forest of just 3 trees (each with max_depth=3) was trained separately to show what the individual trees look like. Each tree was trained on a slightly different random sample of the data, so they make different splits even though they all look at the same features. The three trees below show this variety. When the full 100-tree forest makes a prediction, it is essentially taking a vote across 100 trees like these.

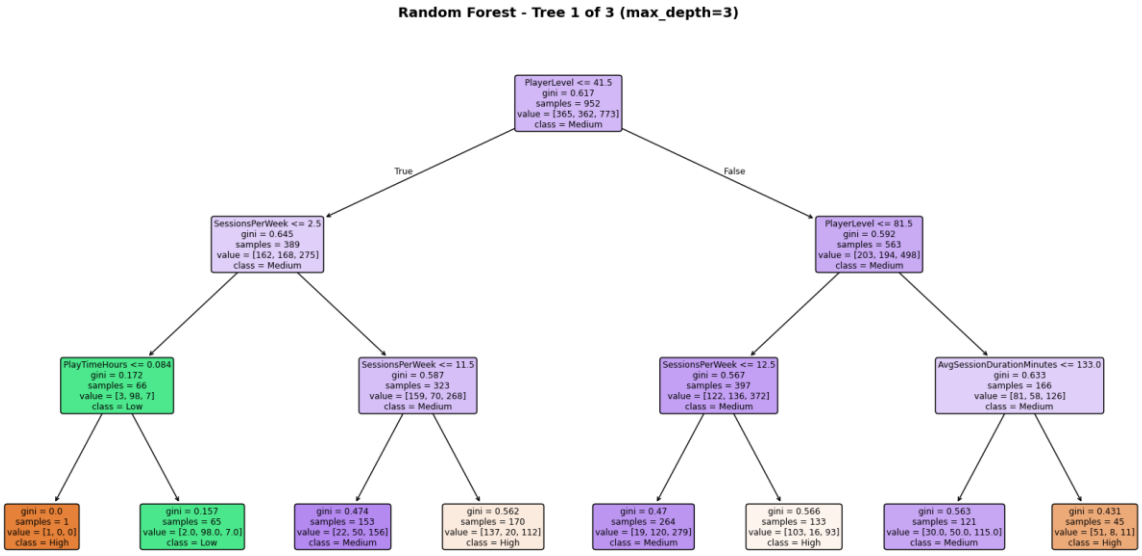


Figure 54: Random Forest Tree 1 of 3 (max_depth=3).

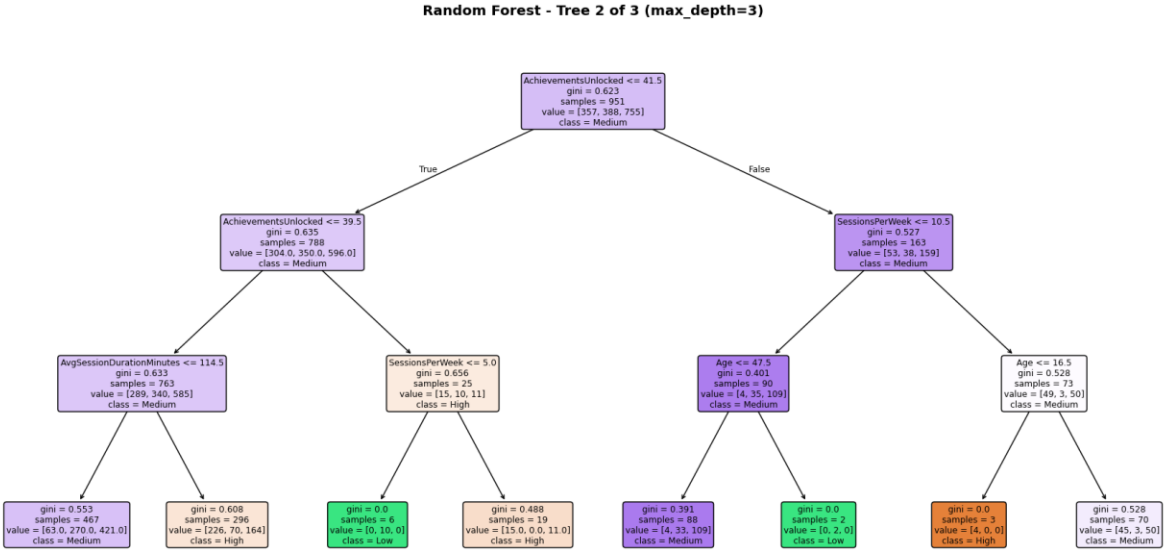


Figure 55: Random Forest Tree 2 of 3 (max_depth=3). Note the different root split compared to Tree 1.

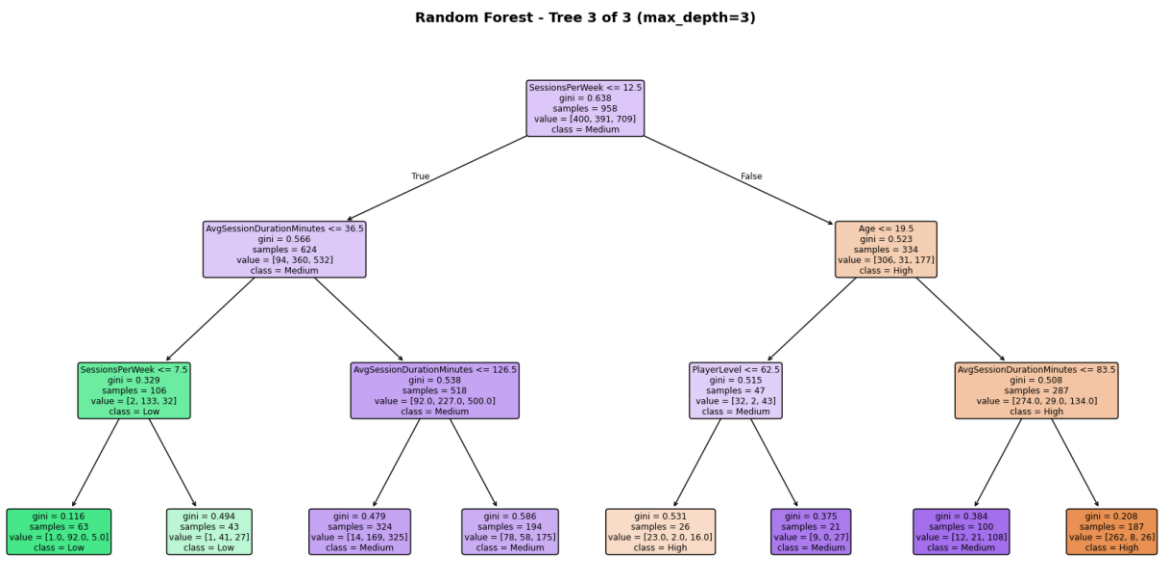


Figure 56: Random Forest Tree 3 of 3 (max_depth=3). Each tree sees different data and splits differently.

AdaBoost

AdaBoost (Adaptive Boosting) takes a different approach from Random Forest. Instead of training many trees independently and voting, it trains models one after another, where each new model focuses on the examples that the previous models got wrong. The mistakes get more weight each round, so the ensemble gradually concentrates on the hard cases. The model here used 100 boosting rounds.

AdaBoost Data Preparation

AdaBoost used the same seven numeric features and the same 75/25 stratified split. Like Random Forest, it is tree-based and does not require feature scaling, so the raw values were used.

AdaBoost Results

AdaBoost reached an accuracy of 0.798, noticeably lower than the Random Forest's 0.850. The confusion matrix shows it did well on Medium but struggled more with the Low and High classes compared to Random Forest. The reason AdaBoost underperformed here is likely that it can be sensitive to noise and overlapping classes. Because it keeps focusing on the hardest-to-classify players (the ones in the fuzzy zone between engagement levels), it can end up over-weighting borderline cases that are genuinely ambiguous.

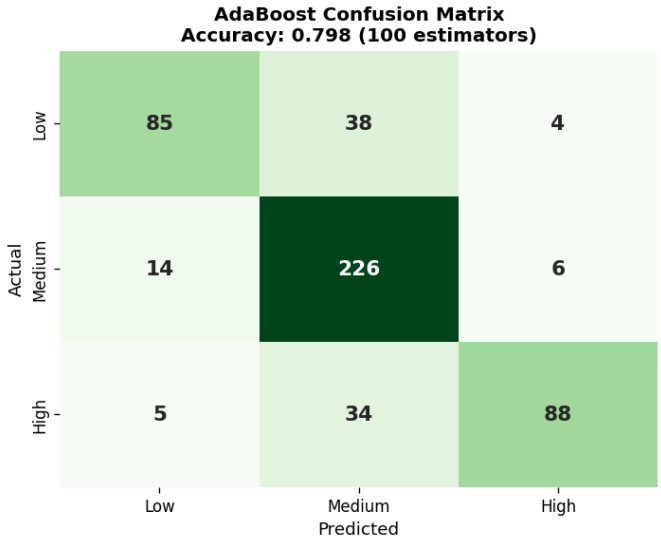


Figure 57: Confusion matrix for AdaBoost (100 estimators, accuracy 0.798).

Comparing AdaBoost to Random Forest directly: Random Forest (0.850) clearly beat AdaBoost (0.798) on this data, a gap of about 5 percentage points. This suggests that for this dataset, the independent-voting strategy of Random Forest handles the overlapping engagement classes better than the sequential error-focusing strategy of AdaBoost.

Stacking

Stacking combines several different types of models and then uses another model (called a meta-learner) to figure out how to best blend their predictions. Unlike Random Forest and AdaBoost, which both build many copies of the same kind of model (decision trees), stacking deliberately mixes different model families so their different strengths can complement each other.

Stacking Data Preparation and Model Selection

Three different base models were chosen to stack: a Random Forest (100 trees), an RBF-kernel SVM, and a Gaussian Naive Bayes classifier. These three were picked because they each approach the problem differently. Random Forest captures rule-based splits, the RBF SVM captures curved boundaries, and Gaussian Naive Bayes captures probability distributions. A Logistic Regression model was used as the meta-learner to combine their outputs. Because the SVM base model needs scaled data, the standardized features were used for the whole stack. Five-fold cross-validation was used during training so the meta-learner could learn how to weight each base model fairly.

Stacking Results

Stacking achieved an accuracy of 0.860, the highest of every model tested in the entire project. The confusion matrix shows the most balanced performance yet, with strong results across all three classes (100 of 127 Low, 232 of 246 Medium, and 98 of 127 High correct). By blending the three different base models, the stack was able to combine their strengths and slightly outperform even the Random Forest on its own.

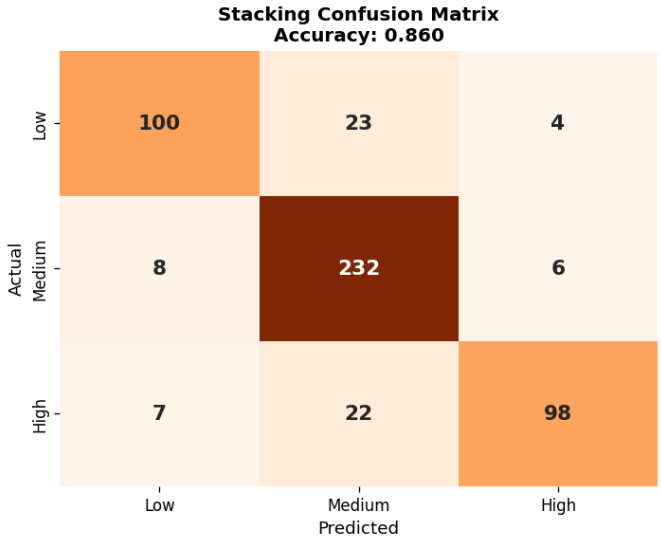


Figure 58: Confusion matrix for the Stacking classifier (RF + RBF SVM + GaussianNB, meta-learner Logistic Regression, accuracy 0.860).

Comparing all three ensemble methods: Stacking (0.860) edged out Random Forest (0.850), and both clearly beat AdaBoost (0.798). The improvement from Random Forest to Stacking was small (about 1 percentage point), which makes sense because Random Forest was already very strong and the extra base models had limited room to add new information. Still, the fact that stacking came out on top shows that combining different model types can squeeze out a little extra accuracy.

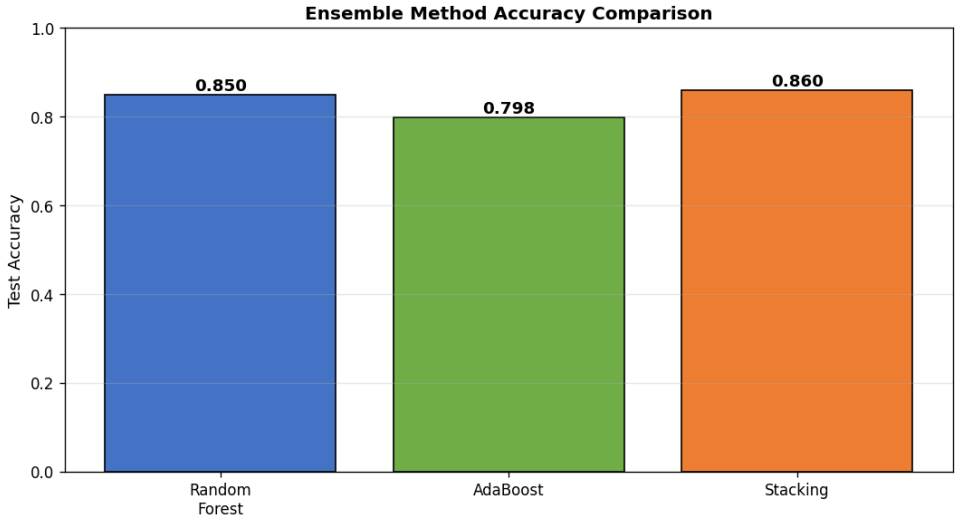


Figure 59: Accuracy comparison across the three ensemble methods. Stacking and Random Forest lead; AdaBoost trails.

Conclusions for Ensemble Methods

The ensemble methods delivered the best accuracy of the whole project, topping out at 86 percent with stacking. The story they tell is consistent with everything that came before: engagement level is genuinely predictable from a player's behavior, and the two features that matter most are how often they play (SessionsPerWeek) and how long their sessions are (AvgSessionDurationMinutes). What the ensembles added was a small but real boost in accuracy by combining many models. Random Forest and Stacking both landed around 85 to 86 percent, showing that pooling predictions beats any single model. The interesting exception was AdaBoost, which actually did worse than Random Forest because it kept obsessing over the genuinely ambiguous borderline players. The overall comparison chart below puts all six models from this module side by side and shows the clear winners.

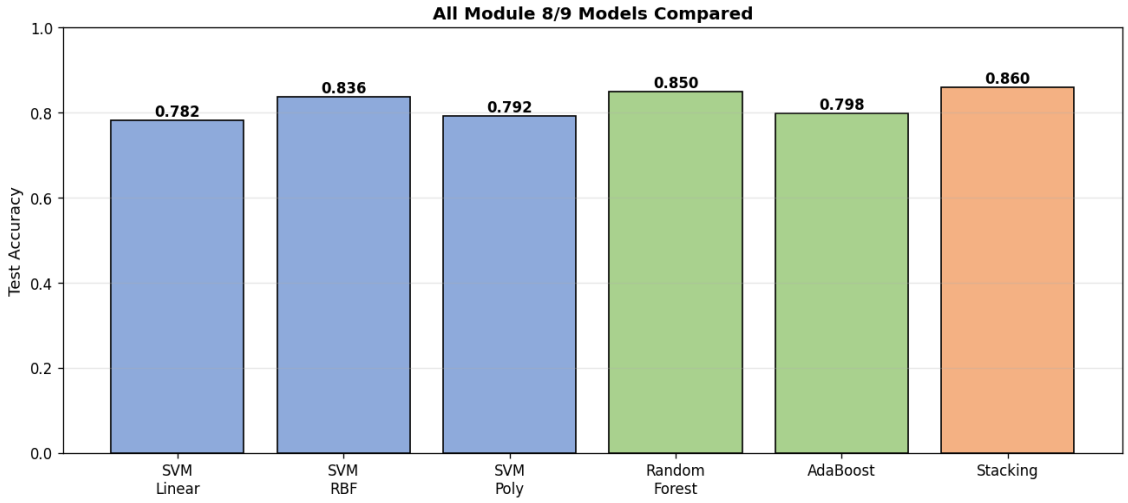


Figure 60: All six Module 8/9 models compared. Stacking and Random Forest lead the pack, with the RBF SVM close behind.

Conclusions

A close look at real data from thousands of gamers has produced a clearer picture of how gaming behavior and mental health actually fit together. Two main story lines emerged from the analysis. The first is that engagement with games, meaning how invested a person becomes in playing, can be predicted with strong accuracy from just a few patterns in how they play. The second is that mental health outcomes like anxiety are far harder to predict from gaming behavior alone, no matter what approach is used. These two findings stand in clear contrast to each other, and that contrast is the most important take-home message of all the analysis. Player behavior follows recognizable

patterns; player feelings do not, at least not from behavior alone. This distinction matters because it pushes back against the common assumption that mental health and gaming are tightly linked in a simple way. The evidence suggests the relationship is much looser and more shaped by individual circumstances than headlines often imply.

Of all the patterns explored, the strongest finding is that pure quantity of gaming is a surprisingly weak signal. Total weekly hours spent gaming had almost no measurable connection to anxiety scores in the survey data. People who played a lot of games were not, on average, more anxious than people who played very little. This conclusion shows up consistently across every angle that was tried, which makes it stand on solid ground. What turned out to matter much more was the shape of the gaming pattern rather than the raw total. Two players who each spent ten hours a week gaming could look very different depending on whether they played in many short sessions or a few long ones. The pattern of engagement, in other words, carries more meaning than the headline number of hours. Anyone trying to understand a gamer's habits should look at how time is spread out rather than just how much time is spent.

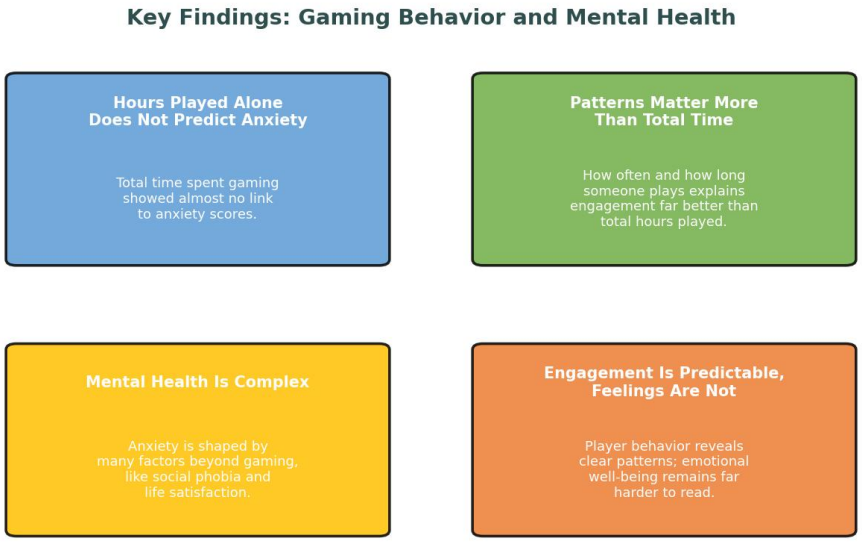


Figure 61: The main findings of the analysis come down to four points about gaming behavior and mental health.

When it comes to predicting how engaged a player is with a game, two simple behaviors stood out above everything else: how often the player logs in each week, and how long the average session lasts. Together these two signals explained most of the difference between casual players and highly engaged ones. Other things that might seem important, like the genre of game being played, the player's age, their gender, where they live, how many achievements they have unlocked, and whether they spend money on in-game purchases, all turned out to matter much less. This finding has

practical value for game designers, parents, and players themselves. If a player suddenly starts logging in much more often, or staying for far longer sessions than usual, those are the signals worth paying attention to. They give a clearer picture of changing engagement than total play time or any single play session ever could. The reason these patterns matter is that they reflect how a player actually relates to a game over time, not just how much time they happen to spend in any given week. A player who logs in briefly but often is sending a very different signal than one who plays a single long marathon and then disappears for weeks, even when the total hours look similar on paper.

Mental health proved to be a different kind of question entirely. Anxiety, social phobia, and life satisfaction are not simple functions of how often a person plays games. The strongest links to anxiety in the data came from other mental health and personality measures, not from gaming habits. Social phobia and life satisfaction explained far more of the variation in anxiety than any gaming-related variable. This lines up with what mental health professionals have said for years: anxiety is shaped by many things, including genetics, life circumstances, work and school stress, family situations, and individual personality. Trying to read a person's mental health from their gaming habits alone is, based on the evidence here, very hard to do well. The flip side of that observation is reassuring in its own way. The number of hours someone games does not, by itself, doom them to anxiety or social problems. Other parts of life carry far more weight.

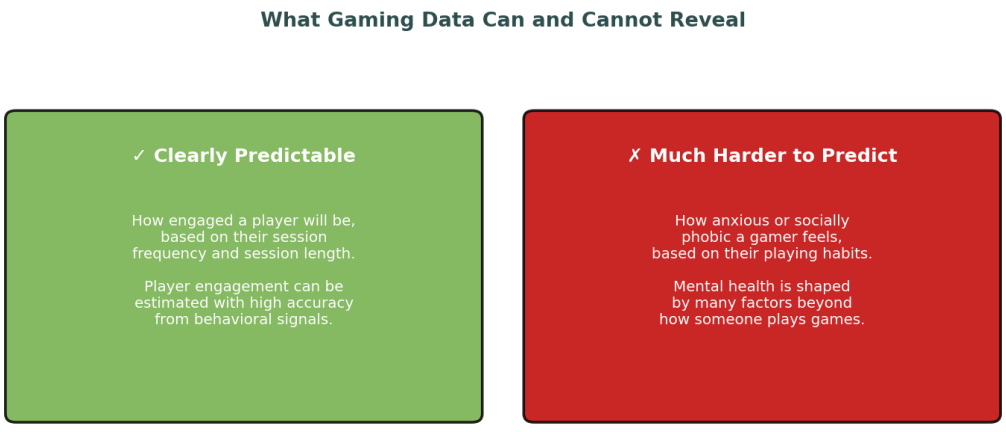


Figure 62: Player behavior can be read from data with reasonable confidence, but mental health remains much harder to predict from playing habits alone.

The broad message that comes out of this work is one of nuance. Gaming and mental health are connected, but the connection is not a simple straight line. Hours of play alone tell very little about how a person feels mentally. What matters more is the

broader picture of how someone lives, what other supports and stressors they have, and how gaming fits into that life. For parents, the takeaway is that watching for changes in patterns is more useful than counting total hours. For schools and healthcare providers, the lesson is that gaming alone is not a reliable warning sign for mental health concerns, and that other factors deserve at least as much attention. For policymakers and developers, the practical message is that designing around session frequency and session length captures more of what actually drives engagement than focusing on total play time. And for individual players, the most useful insight may be the most reassuring: gaming, in moderation and as part of a balanced life, does not appear to be the mental health threat that some headlines suggest. The full story is more interesting, more individual, and ultimately more hopeful than the loudest voices on either side often allow.

About the Author and Improvements

I chose gaming and mental health as the topic for this project because it is something I have a personal stake in. I have spent thousands of hours in games like Final Fantasy XIV, Guild Wars 2, NieR: Automata, and the Persona series, and over the years I have noticed that gaming can be both a source of relaxation and, sometimes, a way to avoid other things I should be doing. I am an introvert, and games have always been my go-to way to wind down after a long day of school or work. I wanted to use the skills I have been learning in this class to see what real data could say about whether the worries about gaming and mental health are backed up by evidence, or whether the picture is more complicated than the headlines make it look. Picking a topic I actually care about made the work much more interesting to do, and it gave me a clearer goal than just running models for the sake of running models.

Working through every step of the data science lifecycle, from gathering and cleaning data all the way through to comparing ensemble models, taught me a lot more than I expected. I came in fairly comfortable with Python and the basic ideas behind machine learning, but I had not really sat down with a single dataset and worked it from end to end before. The cleaning and preparation step turned out to be where most of the real thinking happened. Choosing what to drop, how to handle missing values, and how to engineer labels like the anxiety categories all shaped what the later models could actually find. I also learned to be a lot more careful about how I present results. A confusion matrix with a few specific examples is far more useful than a single accuracy number, and explaining what a model means in plain language is genuinely hard work. By the end of the project, I had a much better sense of what each method is actually good for and where each one tends to fall short.

If I were to do this project again, the first thing I would change is the data itself. The gaming behavior dataset I used was great for engagement classification, but it does not include any mental health measures, which meant I had to keep the two datasets separate and could not directly study how behavior in one connects to mental state in the other. Finding or building a single dataset that captures both gaming patterns and mental health scores for the same players would let the analysis answer much more interesting questions. I would also like to bring in qualitative data, like player interviews or short survey responses, to add context to the numeric patterns. On the modeling side, I would spend more time on hyperparameter tuning. I went with reasonable defaults for most models in this project, but with more time I would use grid search or cross-validation more carefully to squeeze out better performance and to make the comparisons across models more fair. Finally, I would love to extend this work into a longitudinal study that follows the same players over months or years, since most of the

open questions about gaming and mental health are really questions about cause and effect over time rather than snapshots at one moment.

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